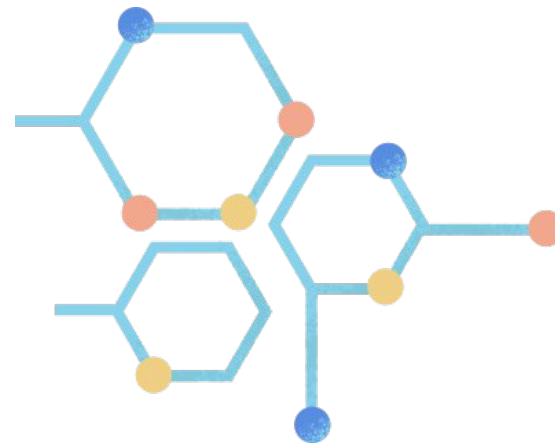




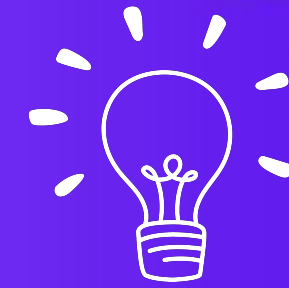
UPSC Prelims

Science & Tech

Class 2

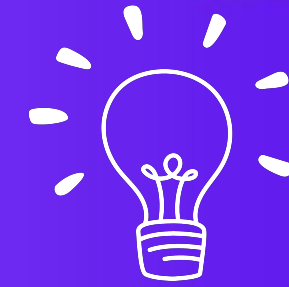
Basic Biology: Cells and Genetics

Cell : Fundamental Unit of Life



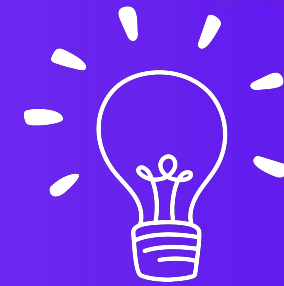
- Unicellular organisms are capable of
 - (i) independent existence and
 - (ii) performing the essential functions of life.
- Anything less than a complete structure of a cell does not ensure independent living. Hence, cell is the fundamental structural and functional unit of all living organisms.
- Anton Von Leeuwenhoek first saw and described a live cell.
- Robert Brown later discovered the nucleus.

Cell Theory



- Schleiden and Schwann, Modified by Virchow
 - (i) all living organisms are composed of cells and products of cells.
 - (ii) all cells arise from pre-existing cells.

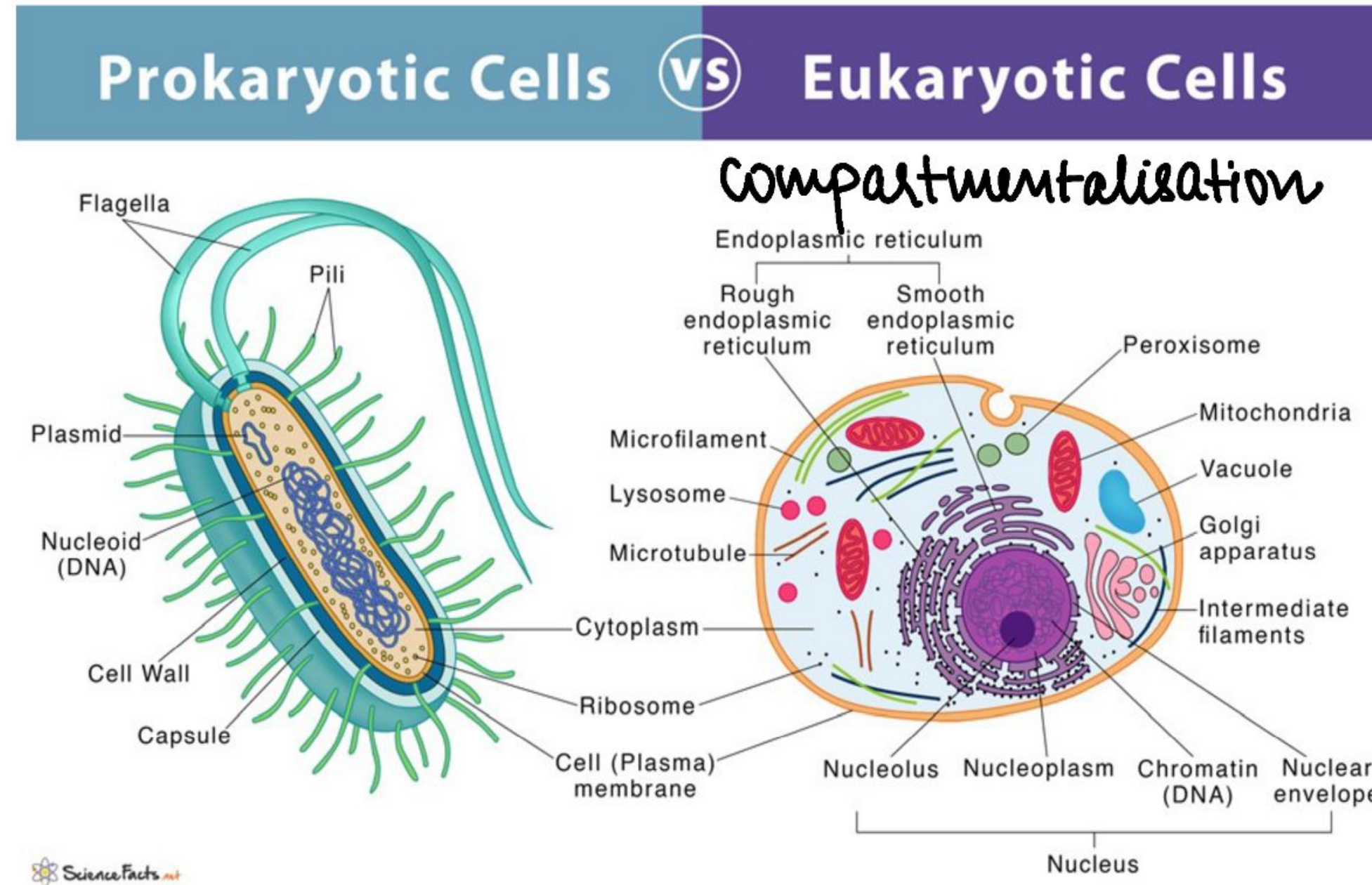
Basic Structure of a Cell



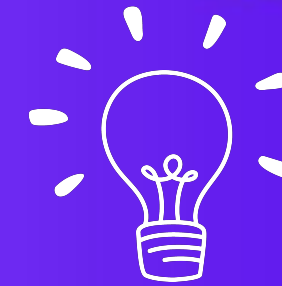
We have already covered this:

- Inside each cell is a dense membrane bound structure called nucleus. This nucleus contains the chromosomes which in turn contain the genetic material, DNA. Cells that have membrane bound nuclei are called eukaryotic whereas cells that lack a membrane bound nucleus are prokaryotic. In both prokaryotic and eukaryotic cells, a semi-fluid matrix called cytoplasm occupies the volume of the cell. The cytoplasm is the main arena of cellular activities in both the plant and animal cells.
- Prokaryotic cells : in monera (will explain later)
- Ribosomes are non-membrane bound organelles found in all cells
- Learn from word breakdown : Pro and Eu + karyon

Eukaryotes vs Prokaryotes



Prokaryotic Cells

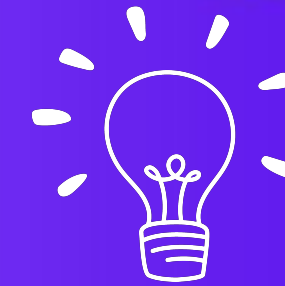


- Bacteria, blue-green algae, mycoplasma and PPLO
 - prokaryotes have a cell wall surrounding the cell membrane except in mycoplasma.
- The fluid matrix filling the cell is the cytoplasm. There is no welldefined nucleus. The genetic material is basically naked, not enveloped by a nuclear membrane. In addition to the genomic DNA (the single chromosome/circular DNA), many bacteria have small circular DNA outside the genomic DNA. These smaller DNA are called plasmids.
- The plasmid DNA confers certain unique phenotypic characters to such bacteria. One such character is resistance to antibiotics.
 - Nuclear membrane only in Eukaryotic cells.

Eukaryotic vs. Prokaryotic Cells

Characteristics	Eukaryotic Cells	Prokaryotic Cells
Definition	Any cell that contains a clearly defined nucleus and membrane bound organelles	Any unicellular organism that does not contain a membrane bound nucleus or organelles
Examples	Animal, plant, fungi, and protist cells	Bacteria and Archaea
Nucleus	Present (membrane bound)	Absent (nucleoid region)
Cell Size	Large (10-100 micrometers)	Small (less than a micrometer to 5 micrometers)
DNA Replication	Highly regulated with selective origins and sequences	Replicates entire genome at once
Organism Type	Usually multicellular	Unicellular
Chromosomes	More than one	One long single loop of DNA and plasmids
Ribosomes	Large	Small
Growth Rate/Generation Time	Slower	Faster
Organelles	Present	Absent
Ability to Store Hereditary Information	All eukaryotes have this ability	All prokaryotes have this ability
Cell Wall	Simple: Present in plants and fungi	Complex: Present in all prokaryotes
Plasma Membrane	Present	Present
Cytoplasm	Present	Present

Eukaryotic Cells



- extensive compartmentalisation of cytoplasm through the presence of membrane bound organelles
- Eukaryotic cells possess an organised nucleus with a nuclear envelope
- Their genetic material is organised into chromosomes.

5 Kingdoms on basis of features:

Monera

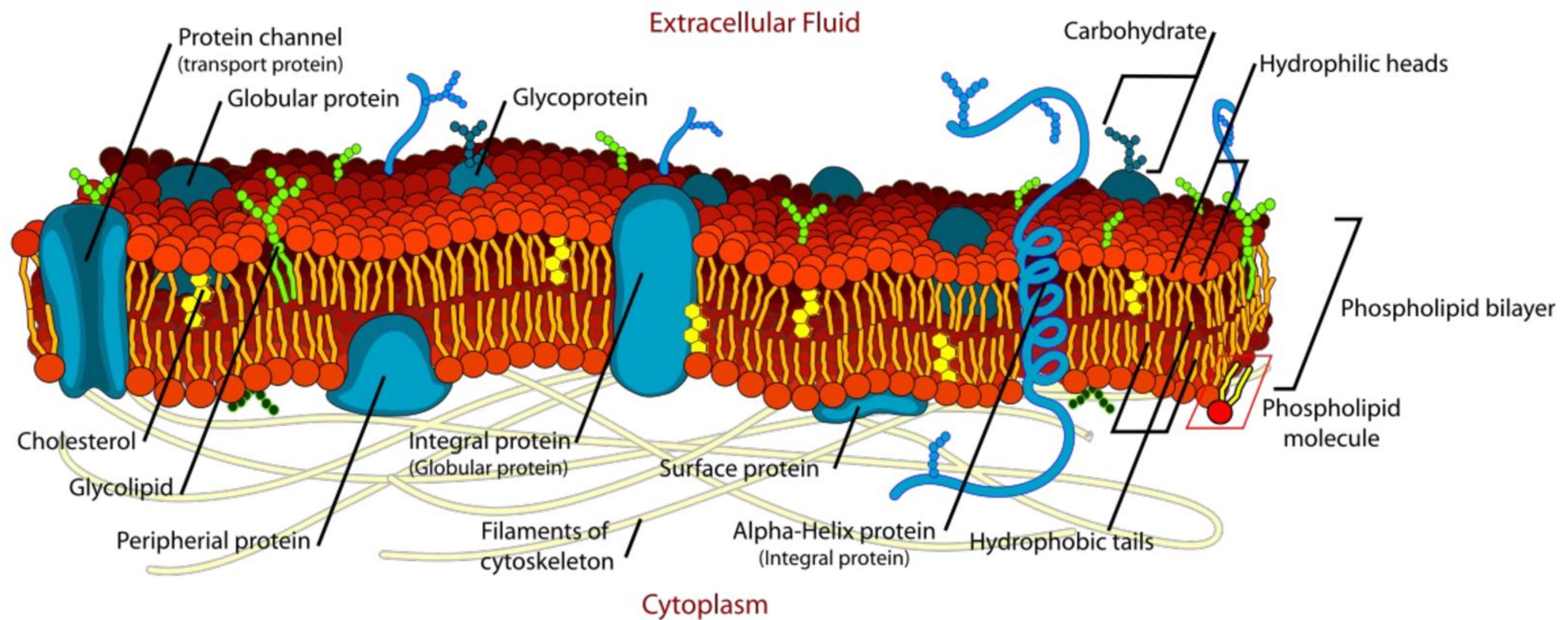
Protista

Fungi

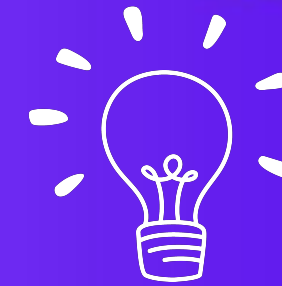
Plantae

Animalia

Cell Membrane : Fluid Mosaic Model



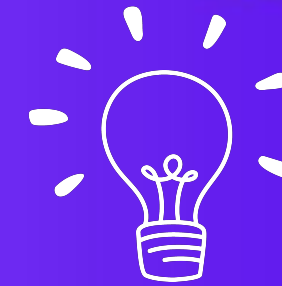
Cell Membrane Continued



mainly composed of lipids and proteins.

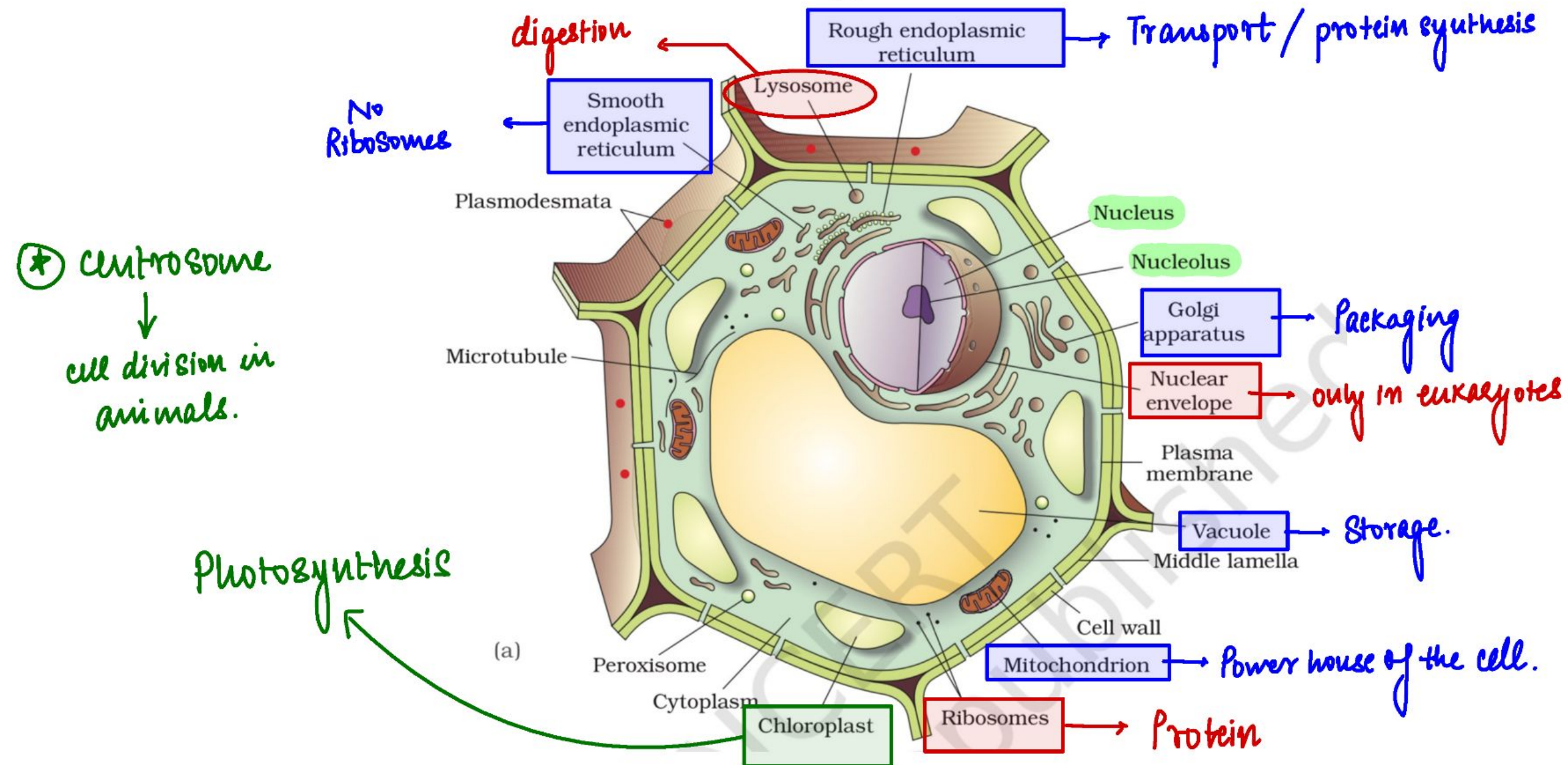
- the major lipids are phospholipids that are arranged in a bilayer.
- Depending on the ease of extraction, membrane proteins can be classified as integral and peripheral. Peripheral proteins lie on the surface of membrane while the integral proteins are partially or totally buried in the membrane.
- quasi-fluid nature of lipid enables lateral movement of proteins : cell growth, formation of intercellular junctions, secretion, endocytosis, cell division etc.
- The membrane is selectively permeable to some molecules present on either side of it.
- Passive transport
- Active transport

Cell Wall



- non-living rigid structure called the cell wall forms an outer covering for the plasma membrane of fungi and plants.
- shape to the cell
- protects the cell from mechanical damage and infection
- cell-to-cell interaction and provides barrier to undesirable macromolecules.
- Algae have cell wall, made of cellulose, galactans, mannans
- Plants : it consists of cellulose, hemicellulose, pectins and proteins.

Cell Organelles



MITOCHONDRIA

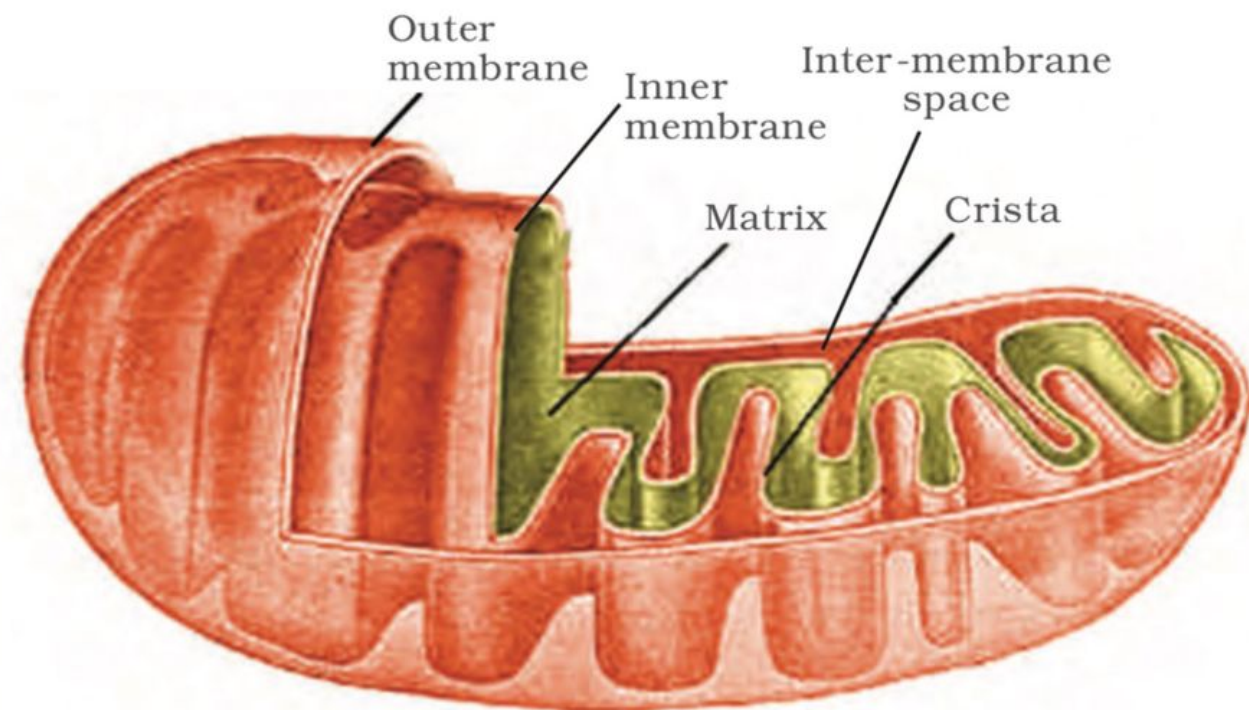
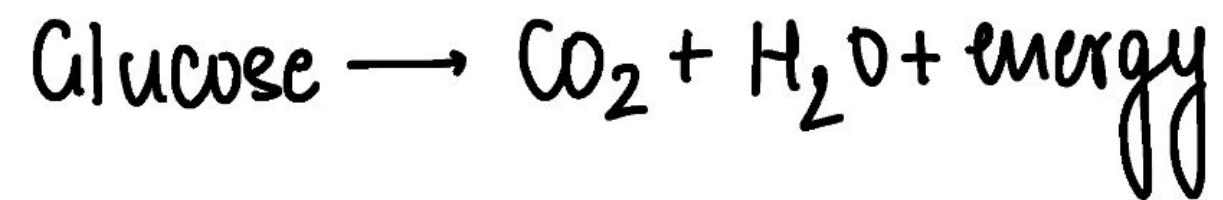
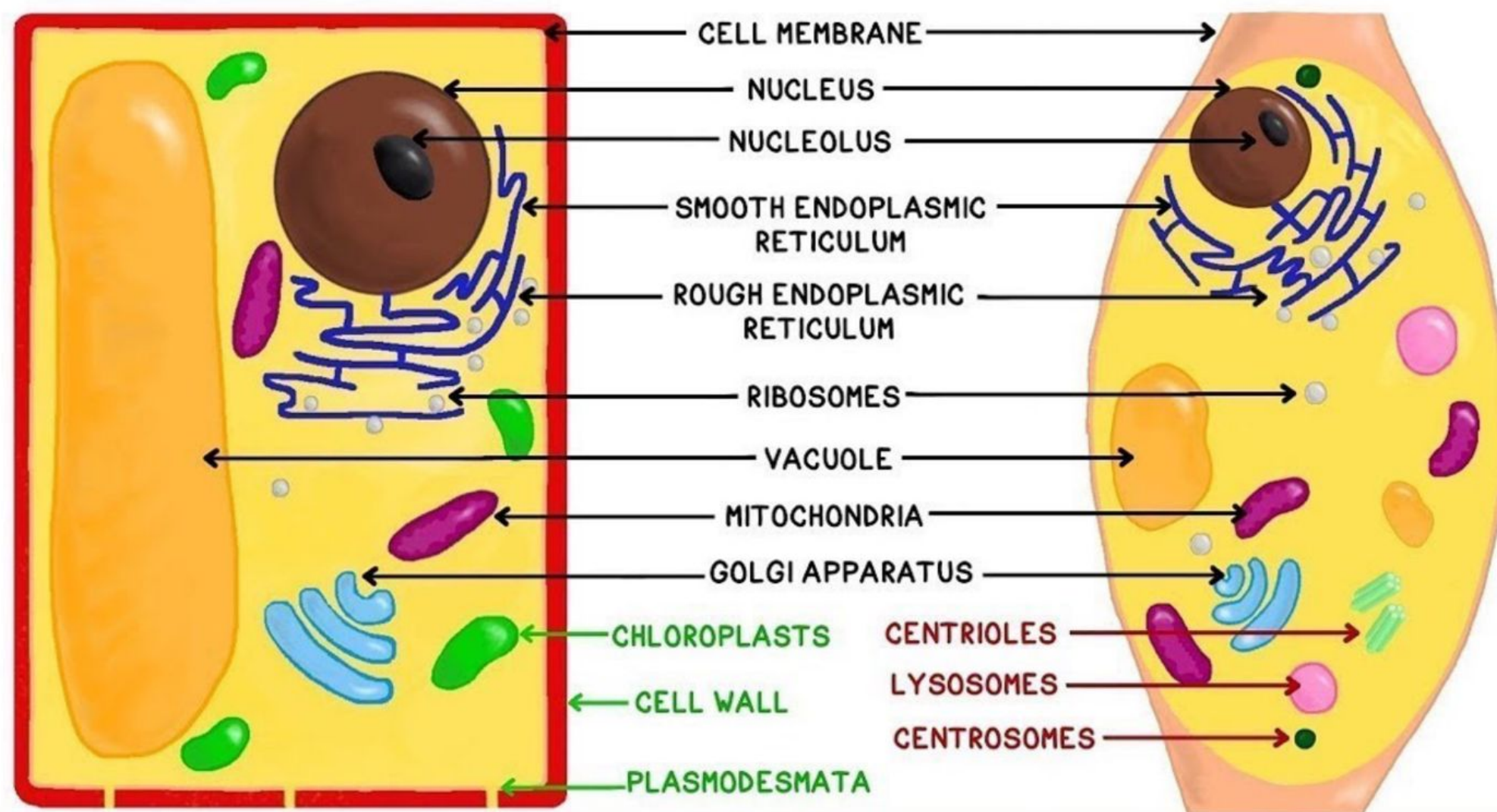


Figure 8.7 Structure of mitochondrion (Longitudinal section)

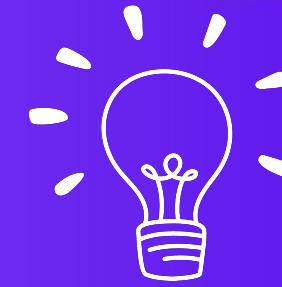


- Power house of cell
 - Produces energy in form of ATP
 - Possesses a single circular DNA molecule.
- [Important]

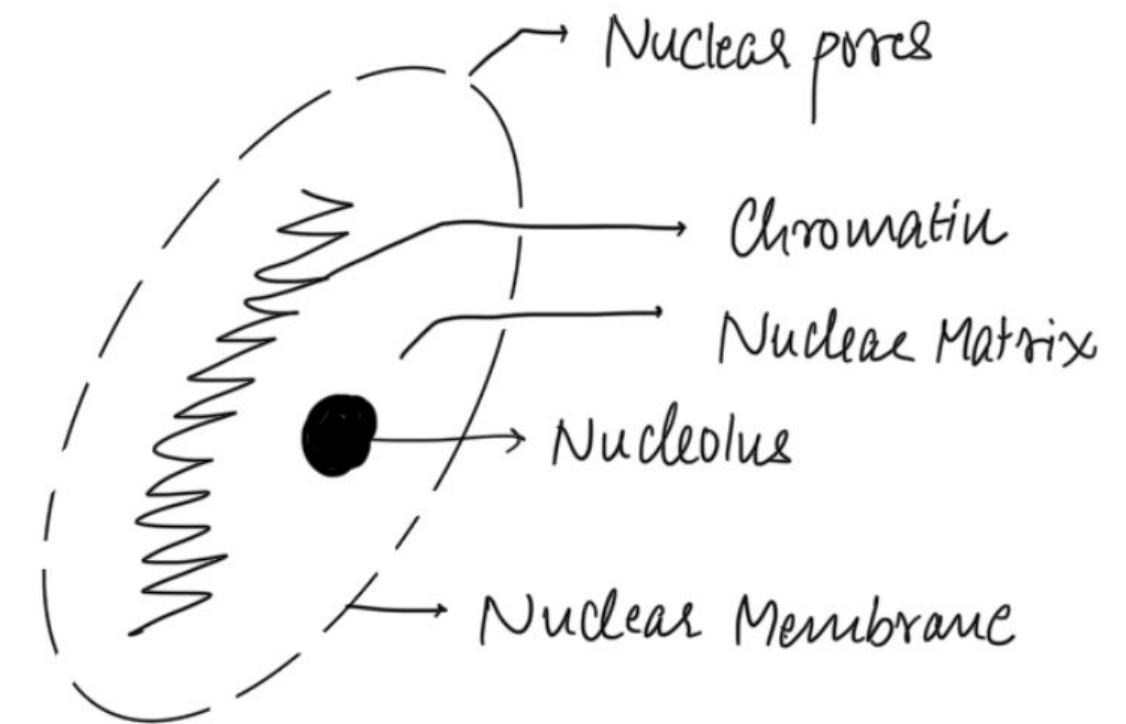
PLANT VS. ANIMAL CELLS

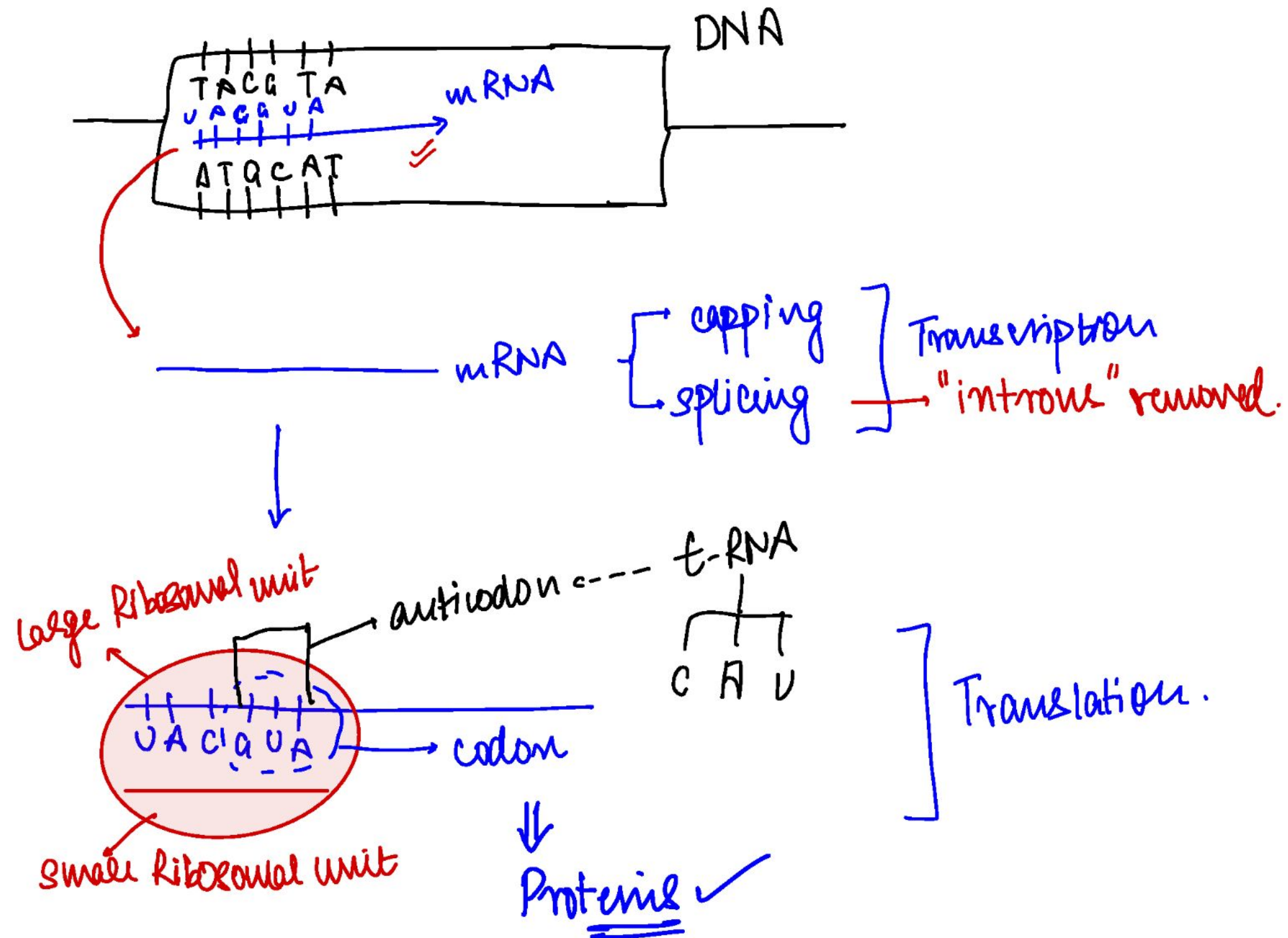


Nucleus



- nuclear pores are the passages through which movement of RNA and protein molecules takes place in both directions between the nucleus and the cytoplasm.
- nuclear matrix or the nucleoplasm contains nucleolus and chromatin.
- The content of nucleolus is continuous with the rest of the nucleoplasm as it is not a membrane bound structure.
- It is a site for active ribosomal RNA synthesis.
- during different stages of cell division, cells show structured chromosomes in place of the nucleus.

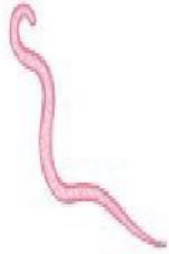




T rna, Mrna, codons and proteins : explain here.

Human Reproduction

- Humans → Made up of cells
- Cells → DNA → Chromatin
Chromosomes.
- DNA → characteristic for a species.
↓
therefore chromosomes are
also characteristic for
a species

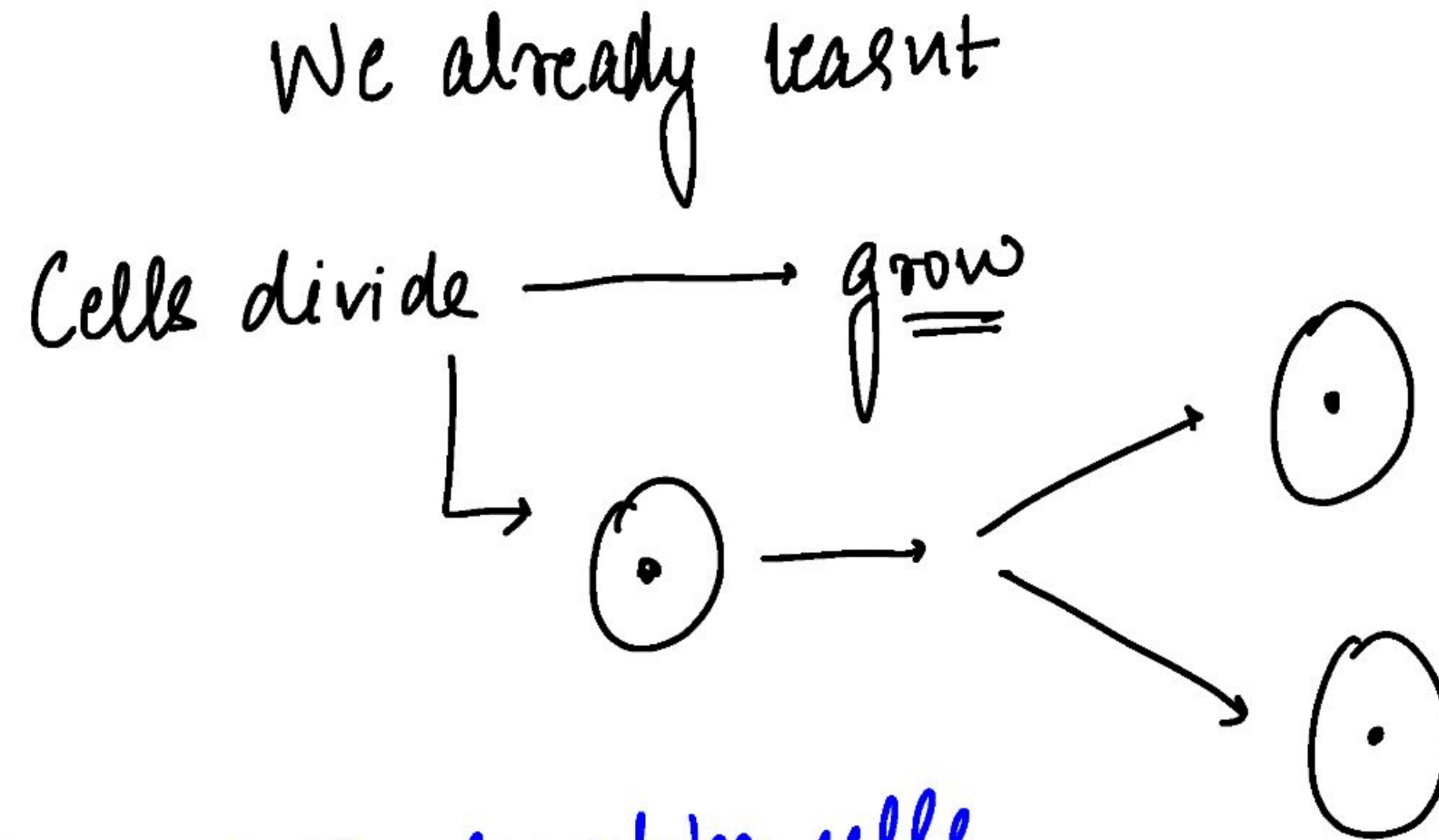
Species	<i>Parascaris equorum</i>	<i>Oryza sativa</i>	<i>Homo sapiens</i>	<i>Pan troglodytes</i>	<i>Canis familiaris</i>
Chromosome #	4	24	46	48	78
Common Name	 Roundworm	 Rice	 Human	 Chimpanzee	 Dog

Chromosomes

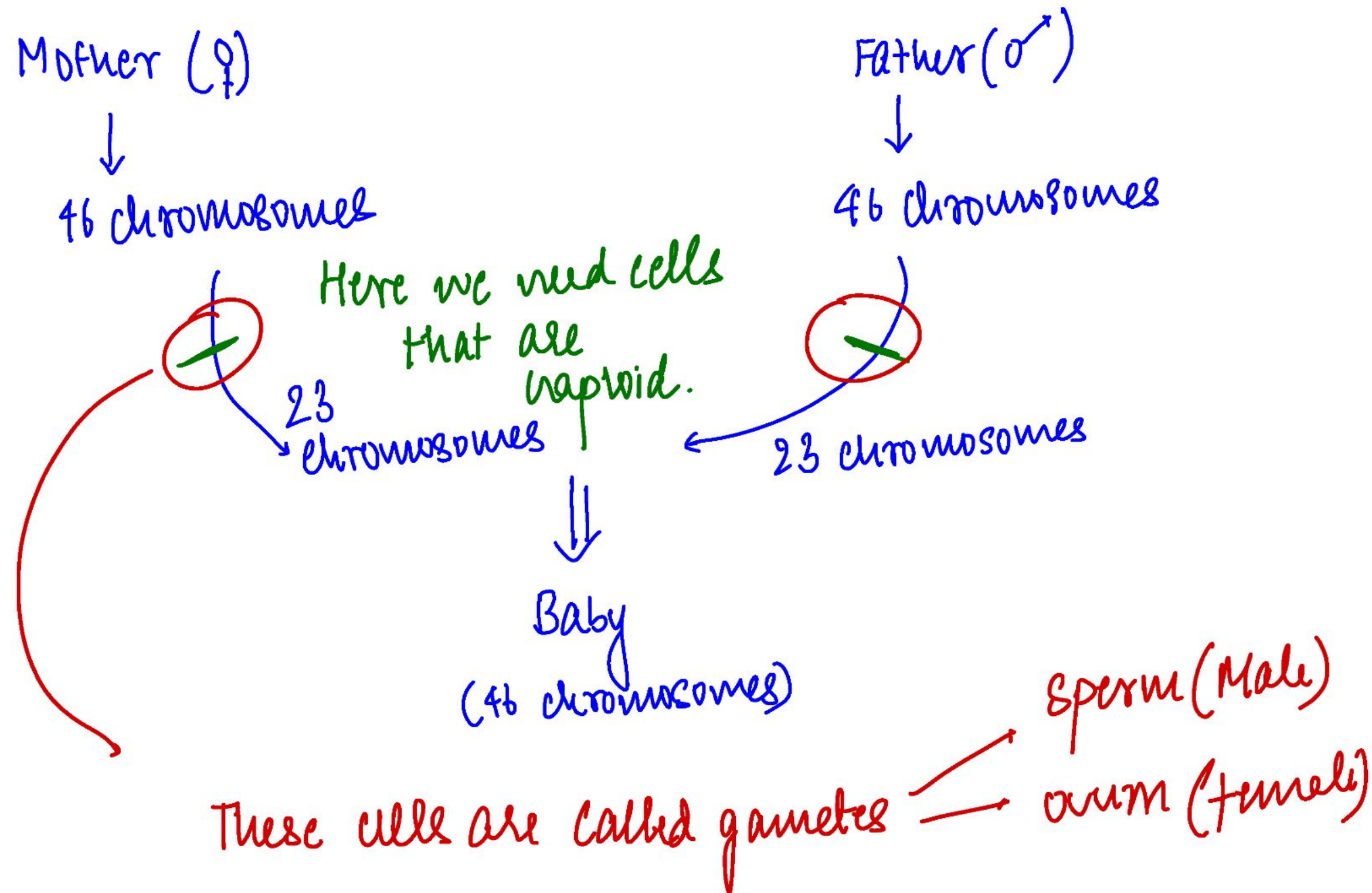
- exist as a pair

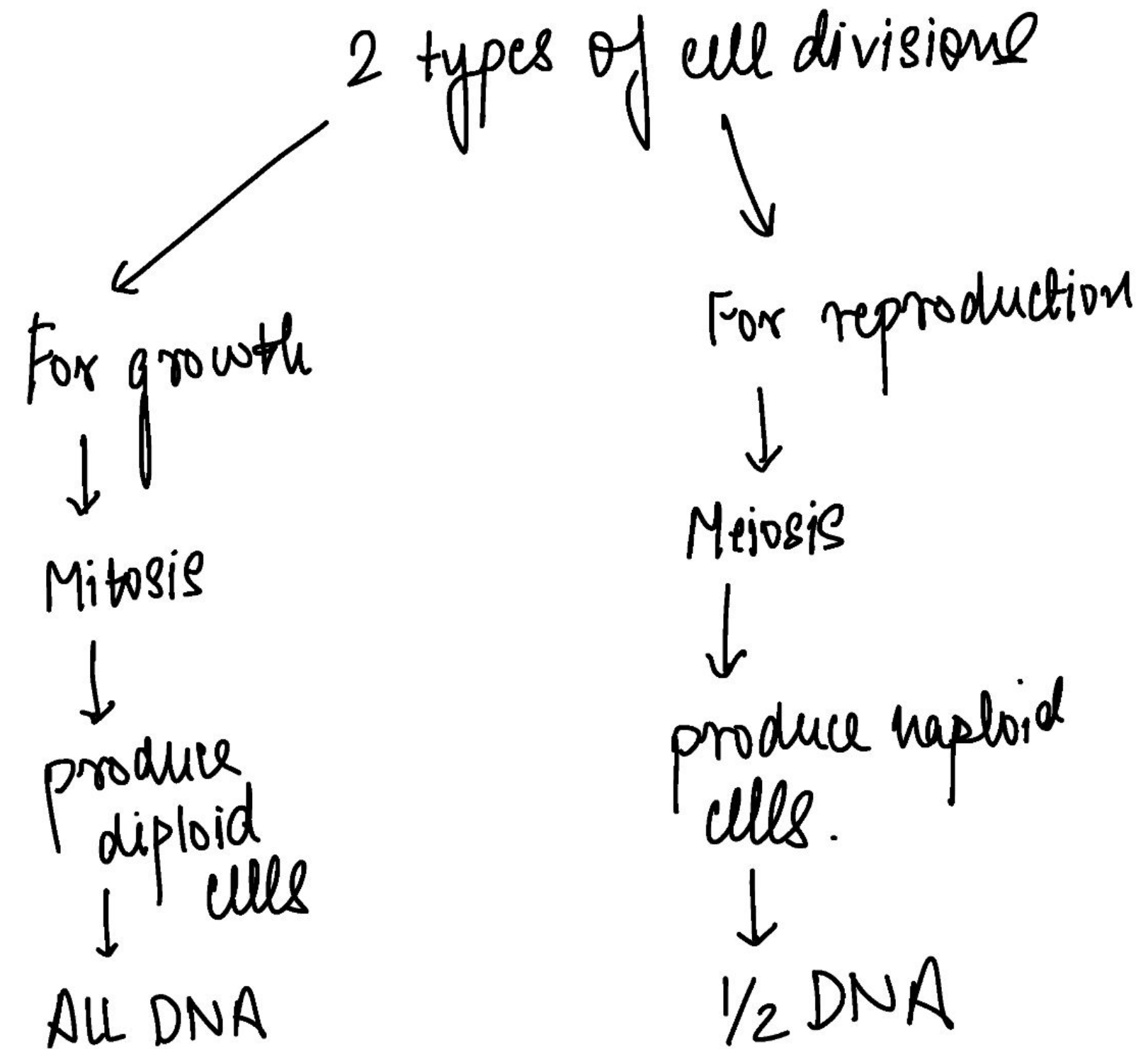
- 46 chromosomes $\rightarrow$ 23 pairs [DNA]

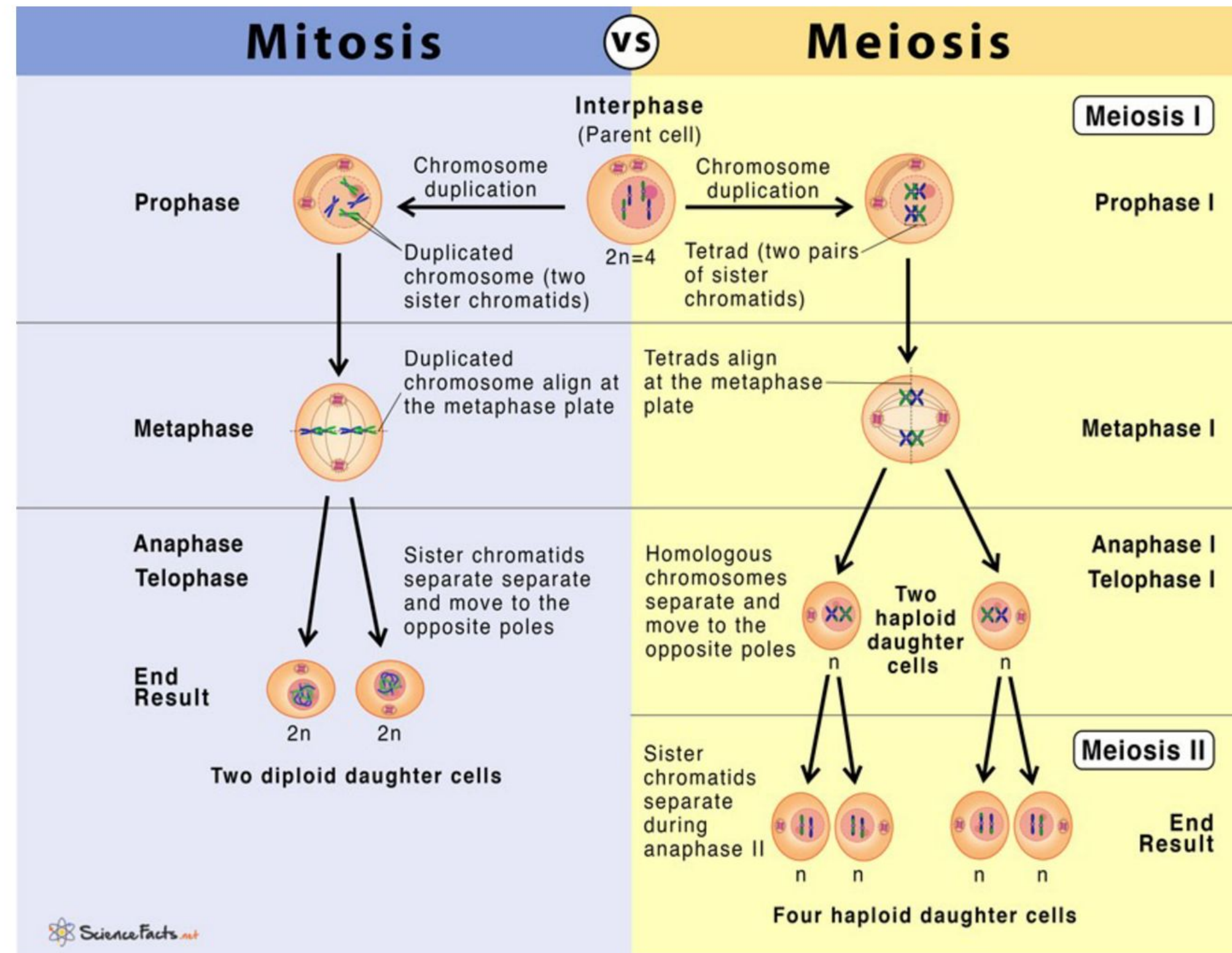
- This is k/a diploid
 - $\swarrow$
Represented as $2(n)$
where $(n) = 23$
- $\rightarrow$ if a cell has only one set
 - $\downarrow$
called haploid (n)
 - $\downarrow$
Why?



Here, both daughter cells
must have same no. of chromosomes
as parent.



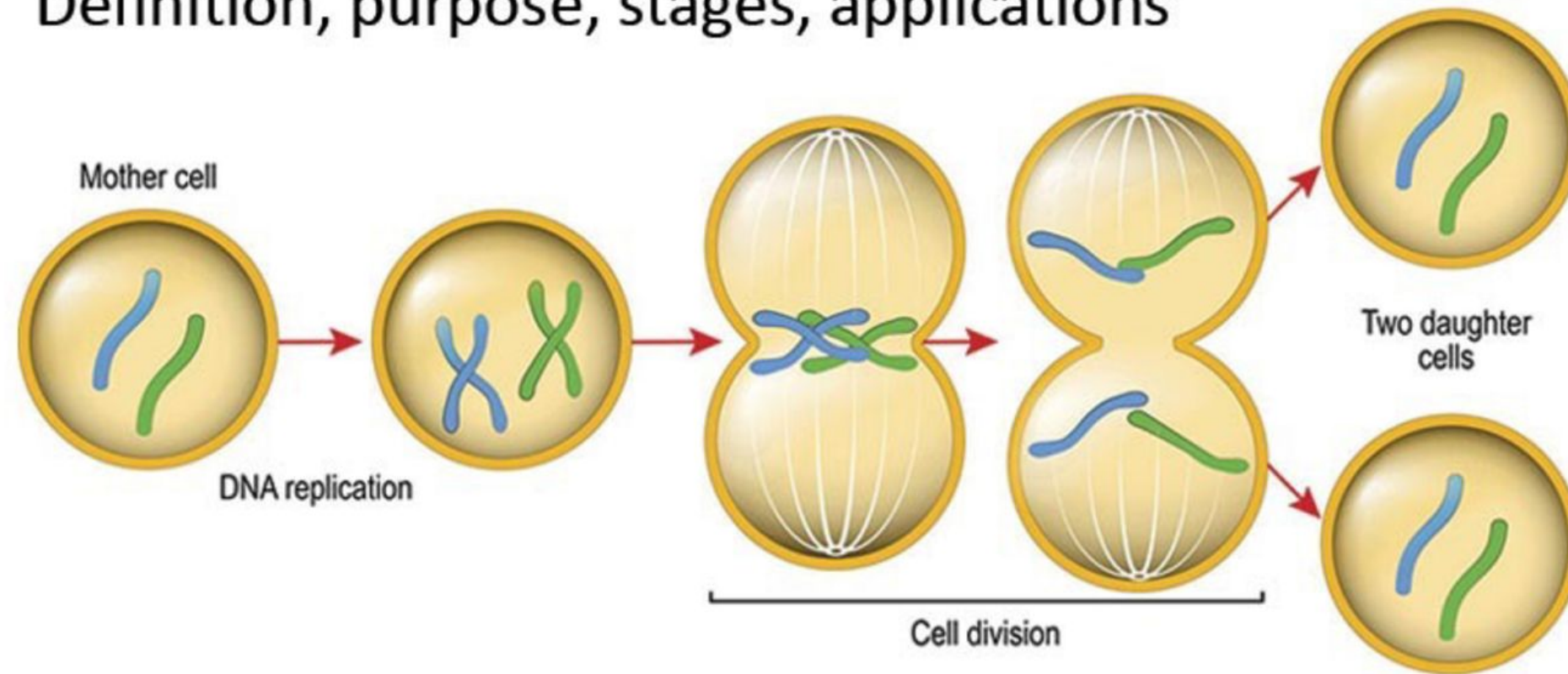




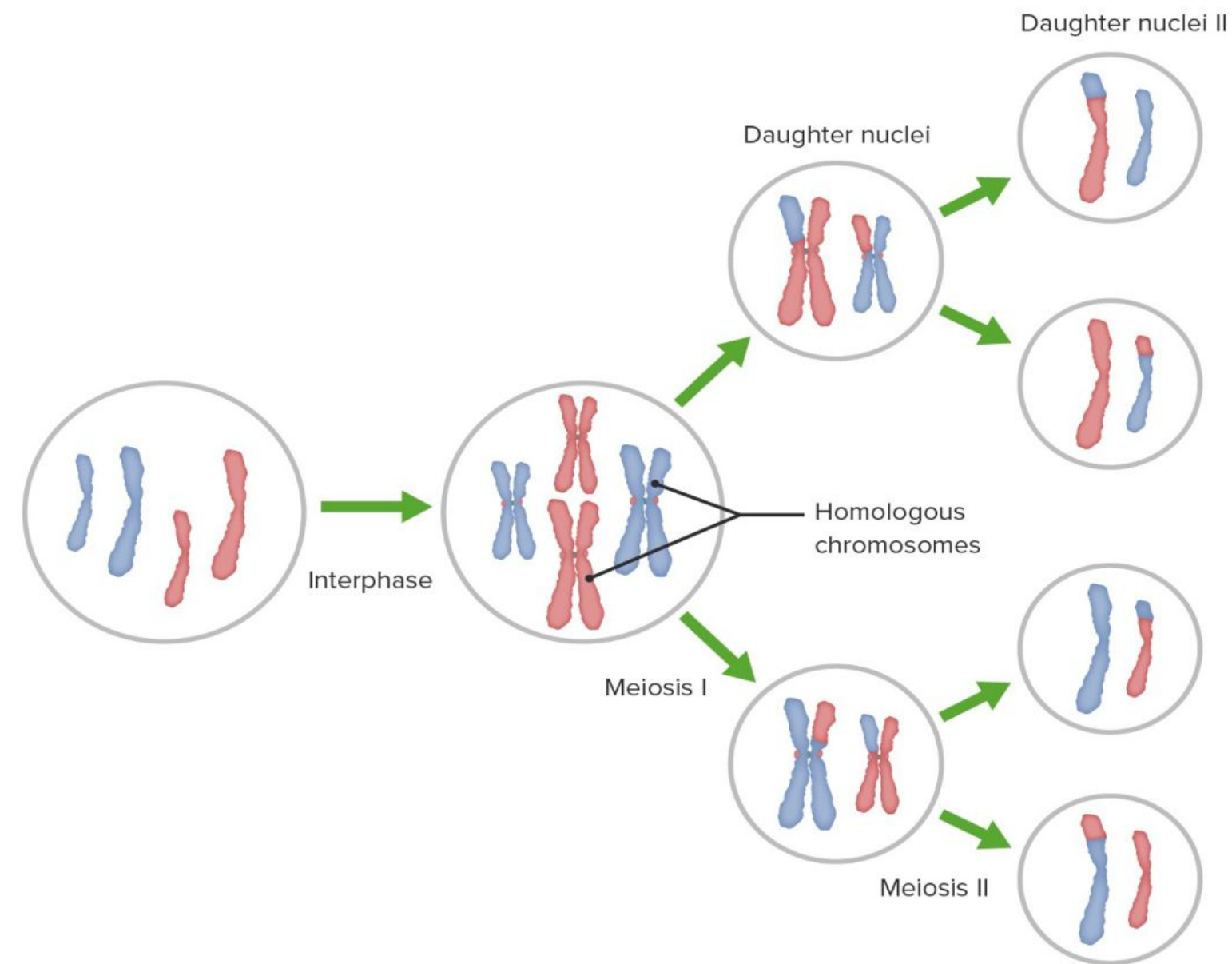
Mitosis

Mitosis

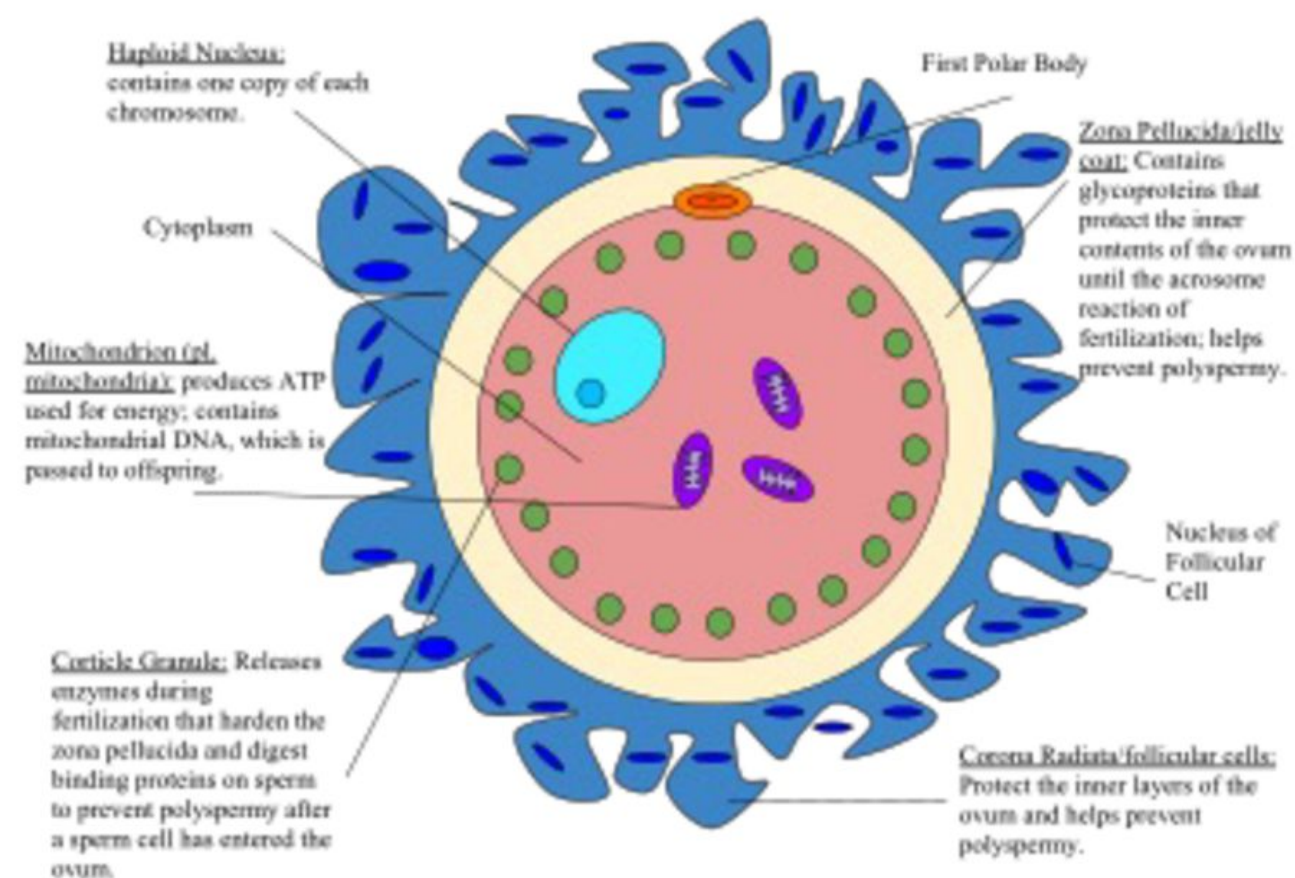
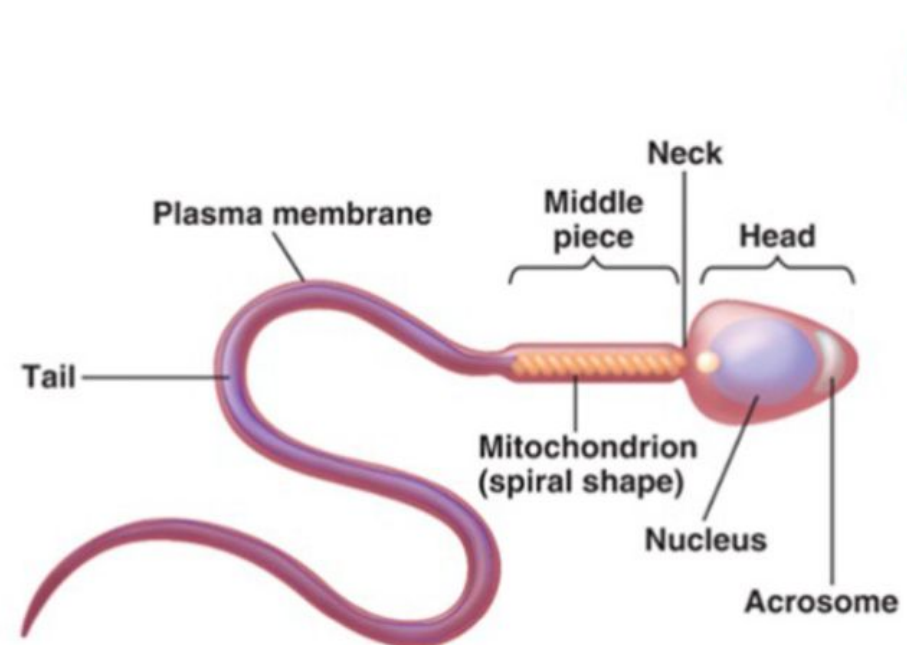
Definition, purpose, stages, applications



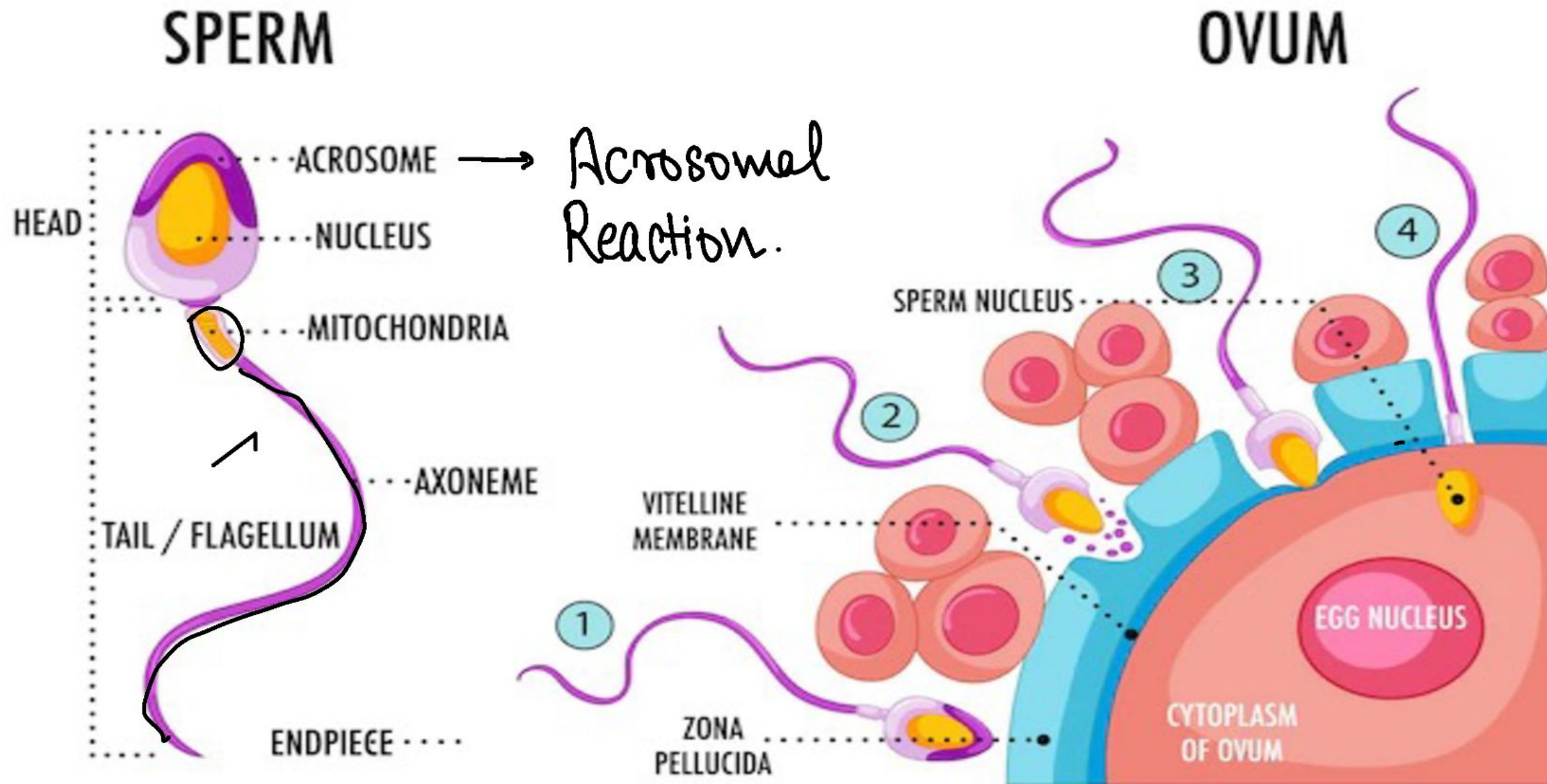
Meiosis



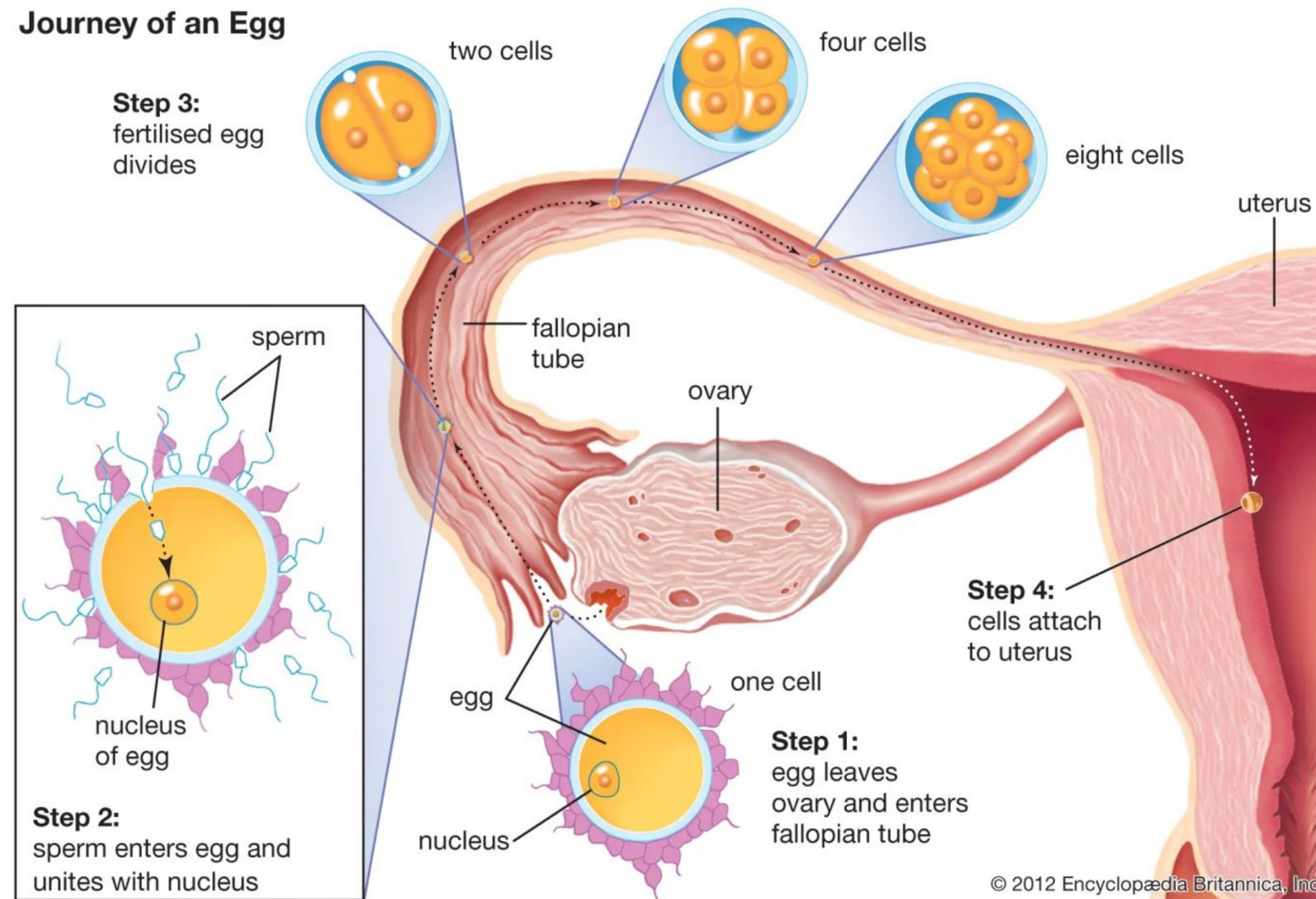
Gametes : Sperm and Ovum



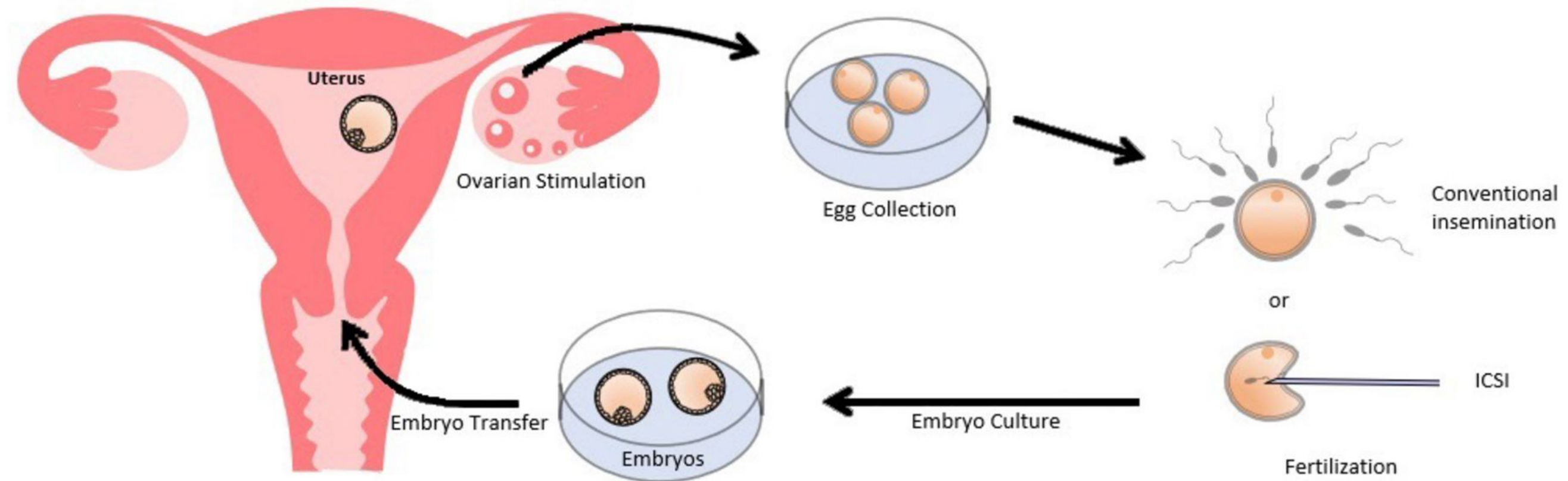
FERTILIZATION



Normal Fertilisation

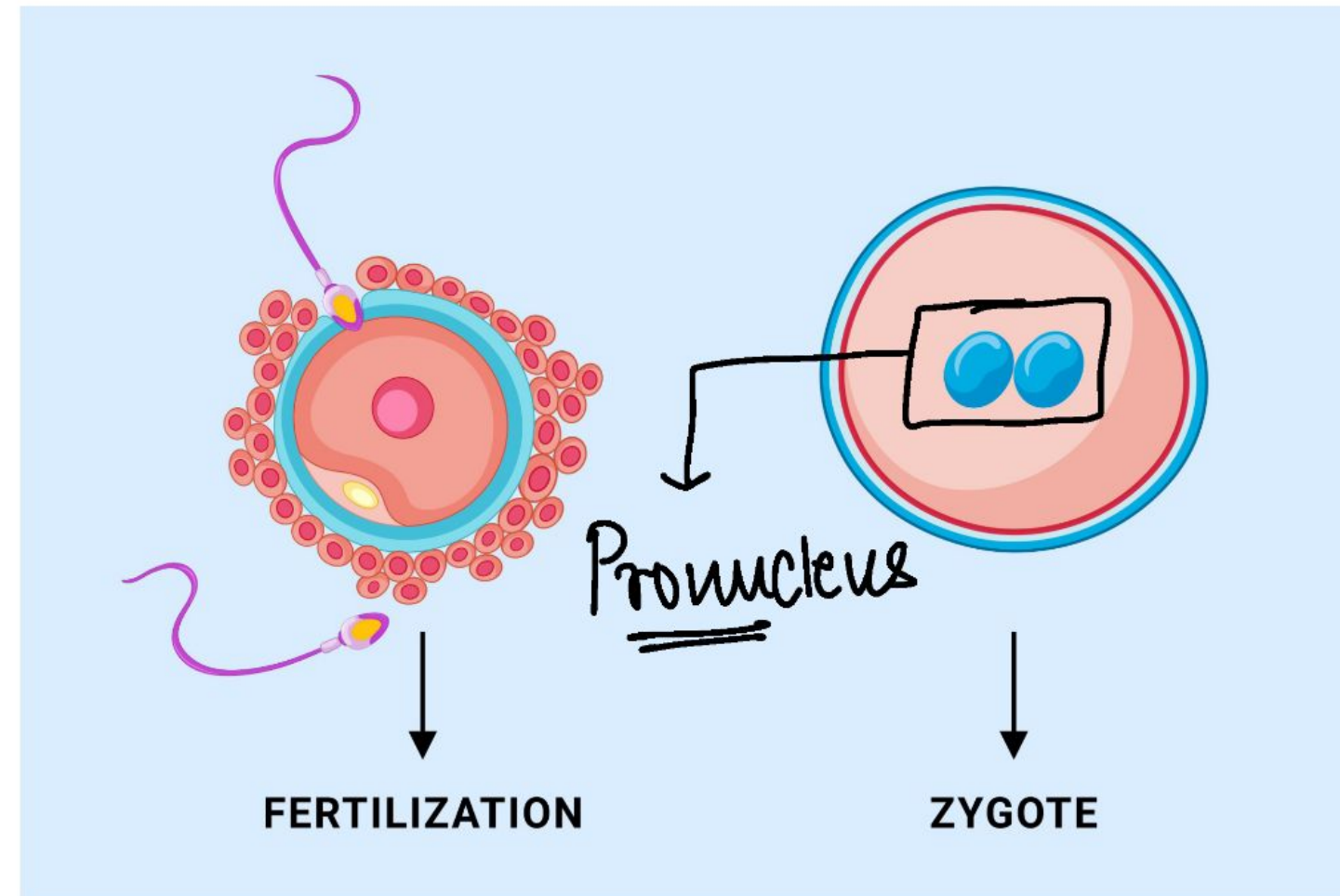


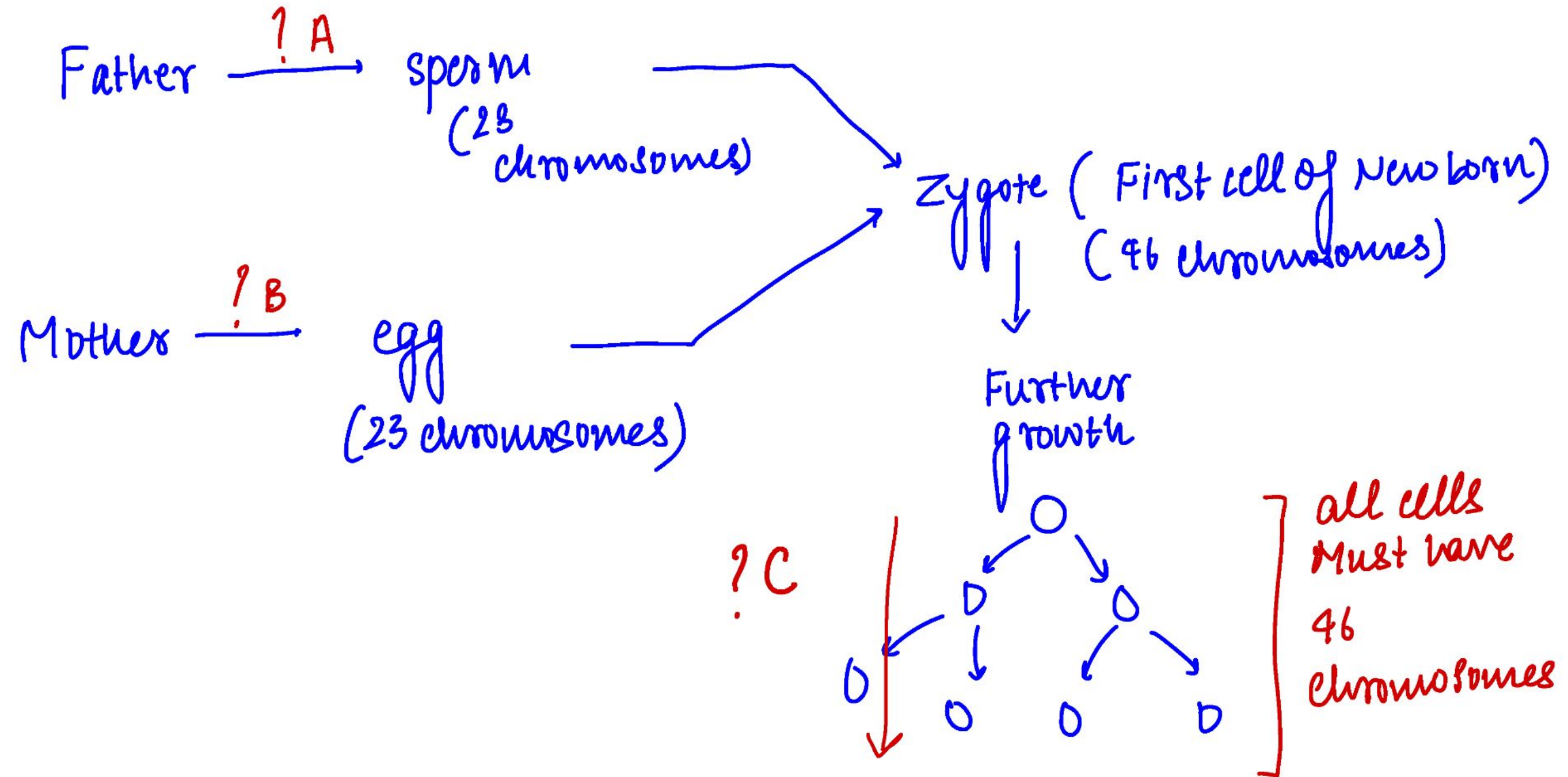
In vitro fertilisation



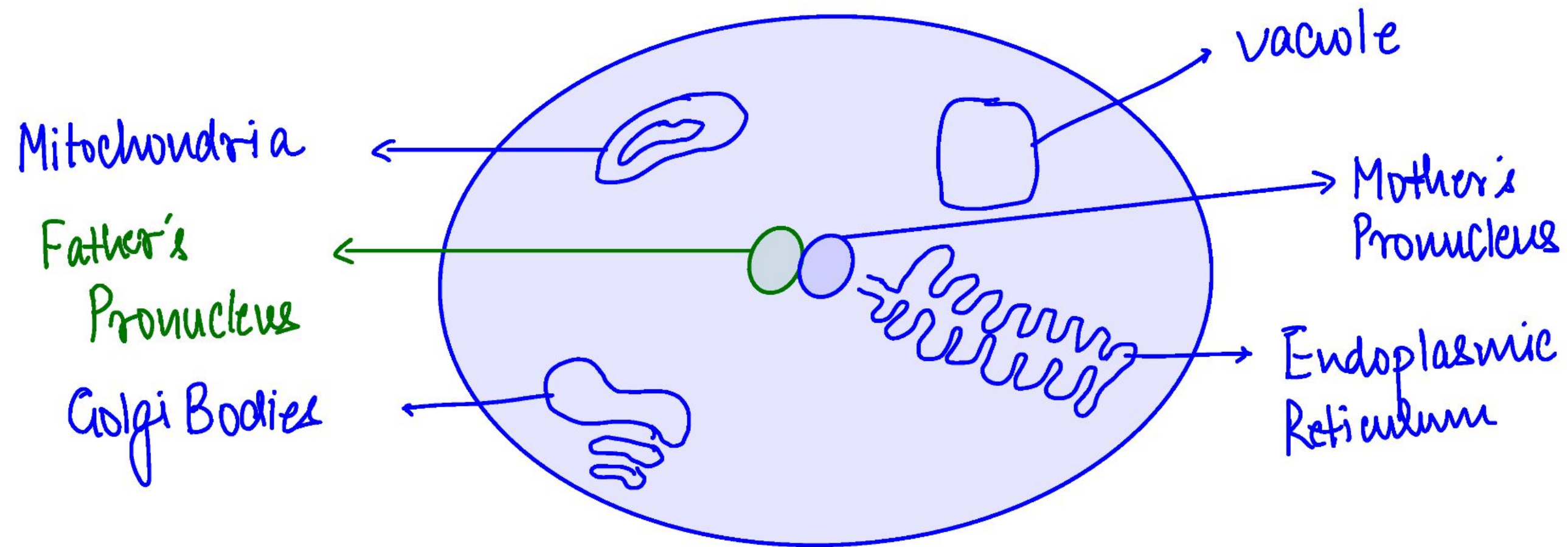
Zygote features

- It has nuclei containing chromosomes from both parents.
- It has organelles only from the mother (ovum)
- Is there any other place other than the nucleus where DNA is present? (try and remember)

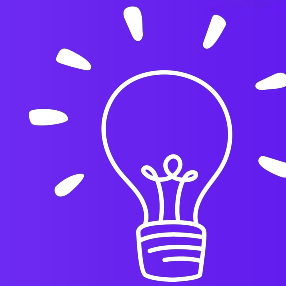




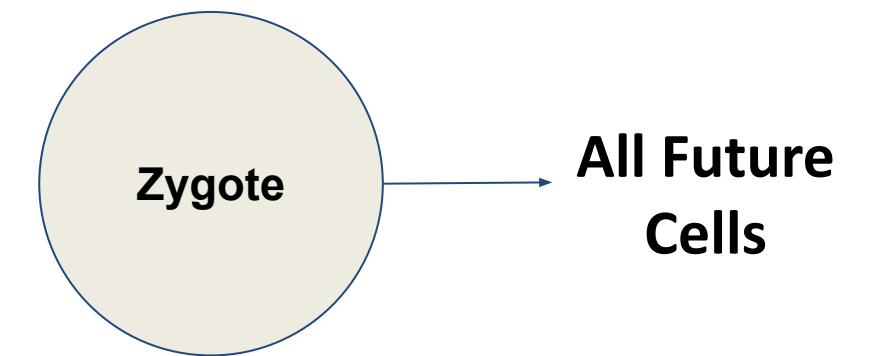
Components of zygote



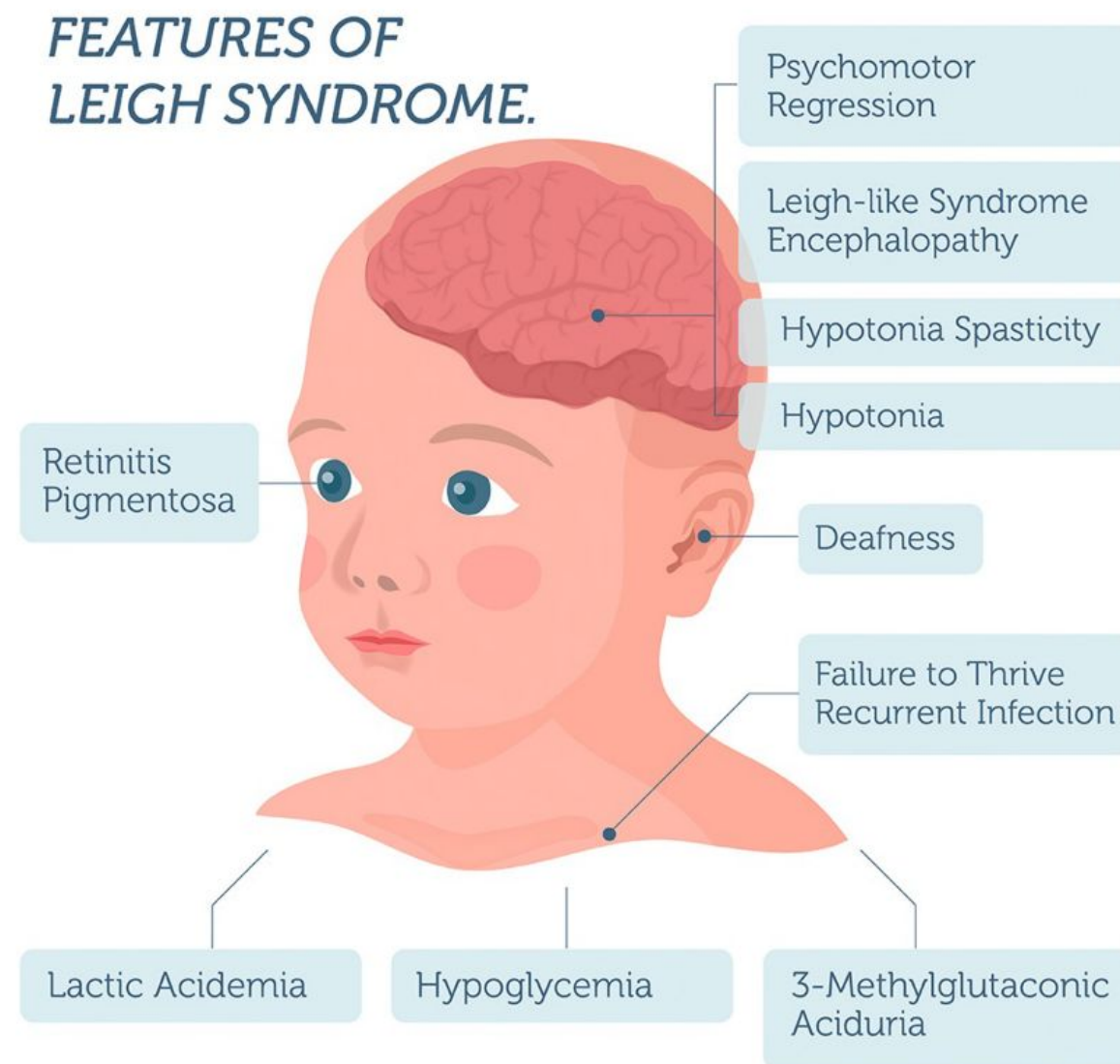
What do we do if a mother's mitochondria is defective?



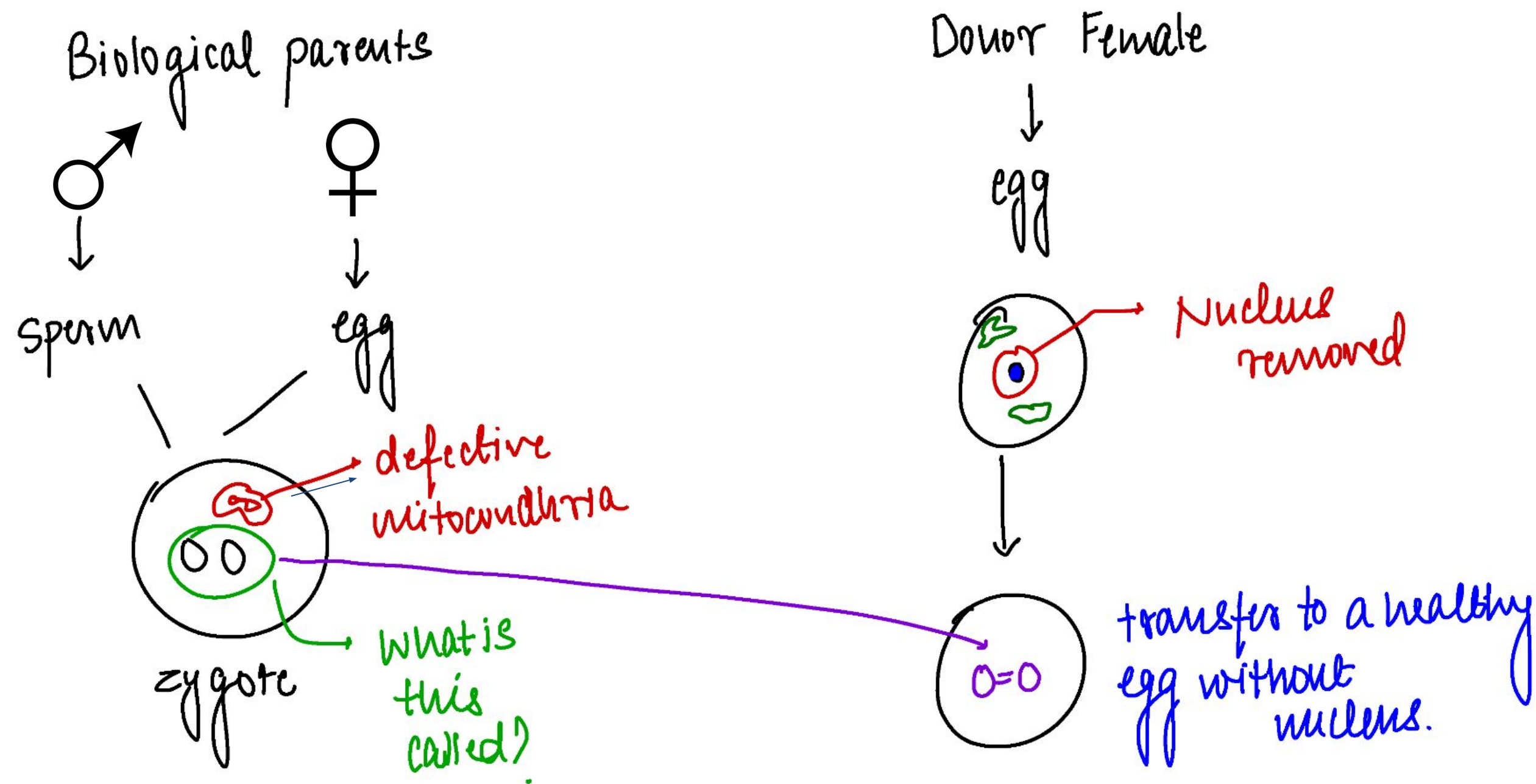
- Replace the mitochondria with healthy mitochondria
- Mitochondrial replacement therapy (MRT), aka mitochondrial donation, is the replacement of mitochondria in one or more cells to prevent or ameliorate disease. MRT originated as a special form of in vitro fertilization in which some or all of the future baby's mitochondrial DNA (mtDNA) comes from a third party. This technique is used in cases when mothers carry genes for mitochondrial diseases.



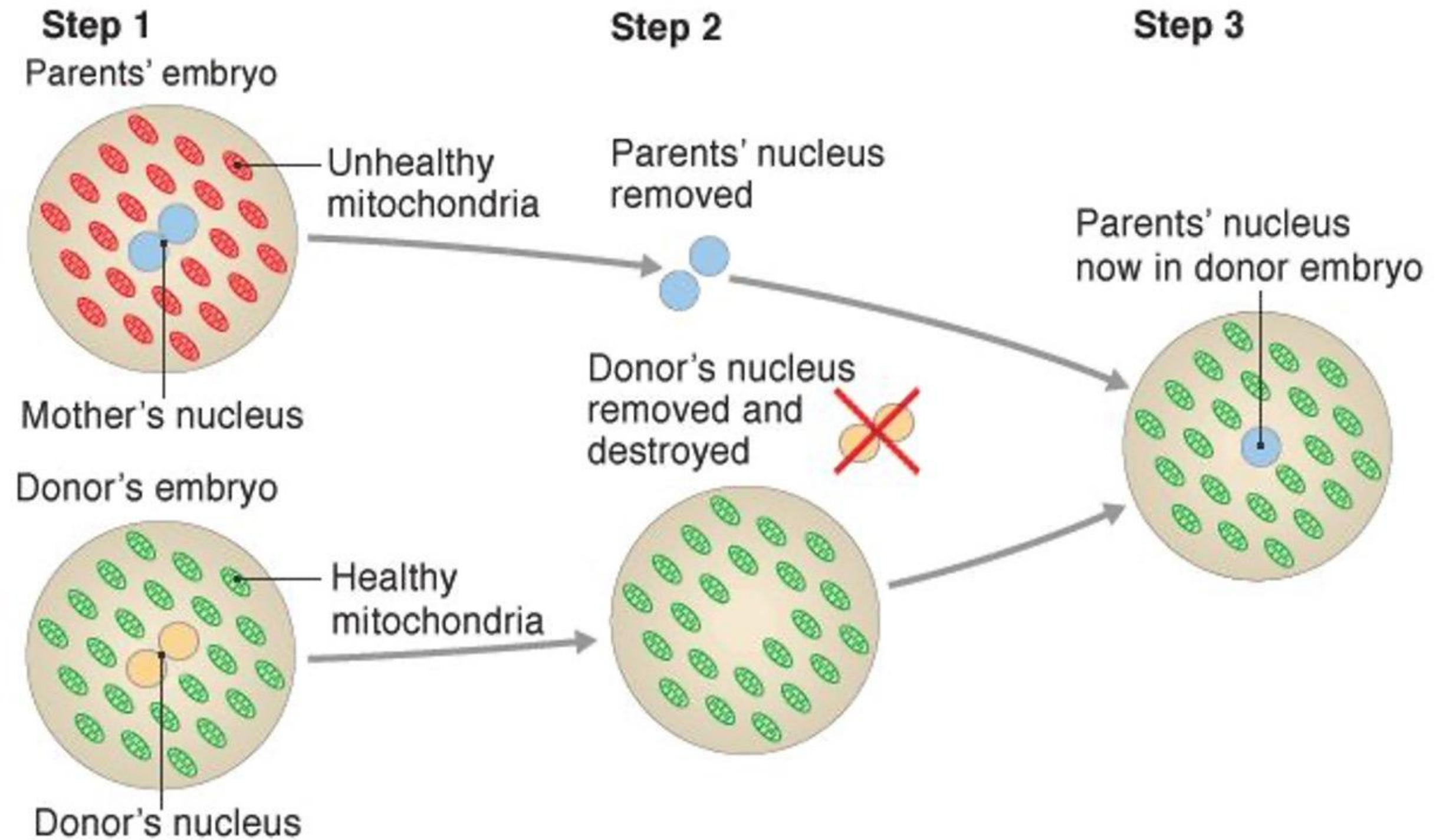
- We need to replace mother's mitochondria with a healthy mitochondria
- Defective mitochondria in mother can cause a disease
↓
eg: LEIGH'S disease



Method-1



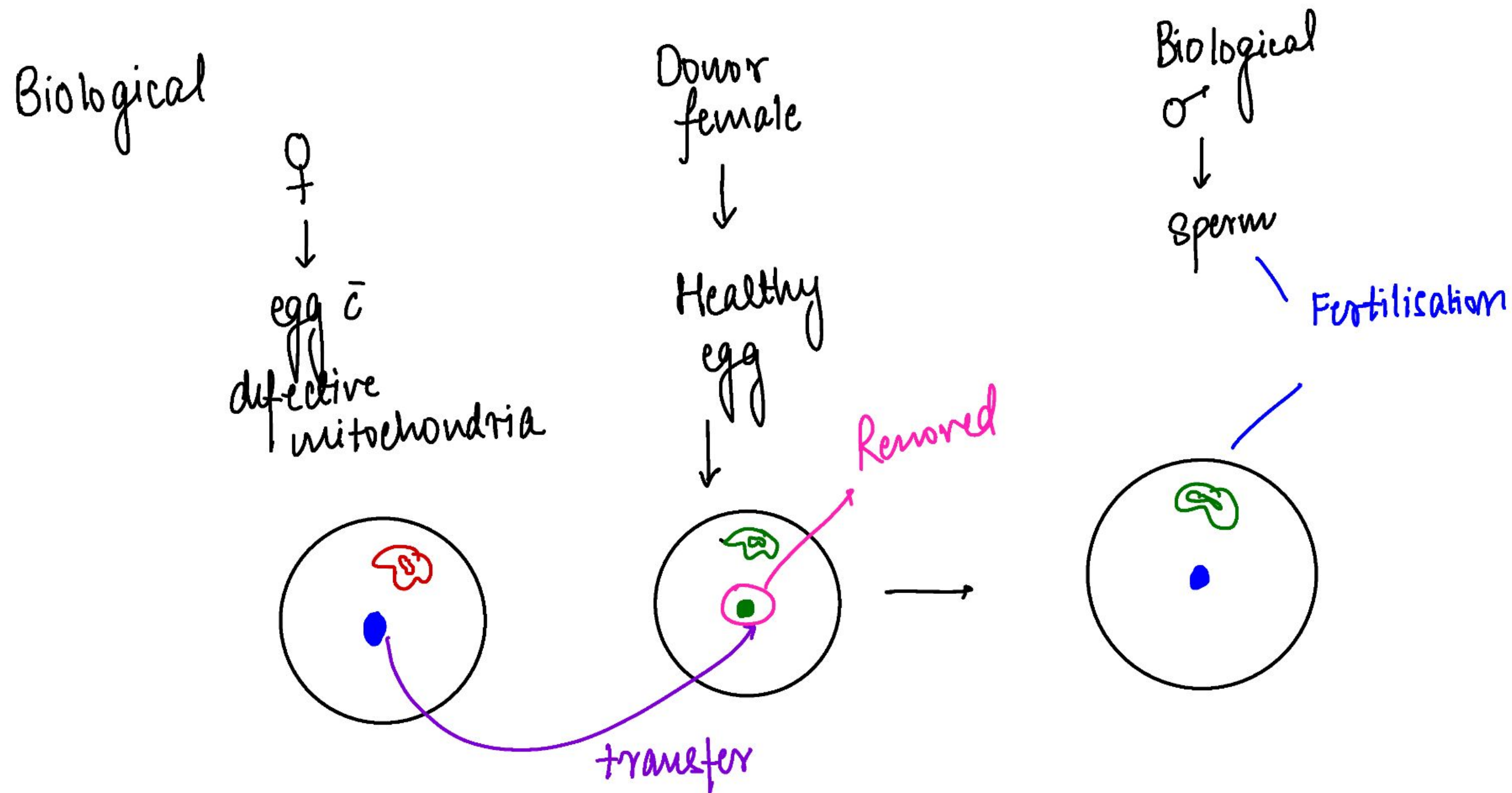
How pronuclear transfer works

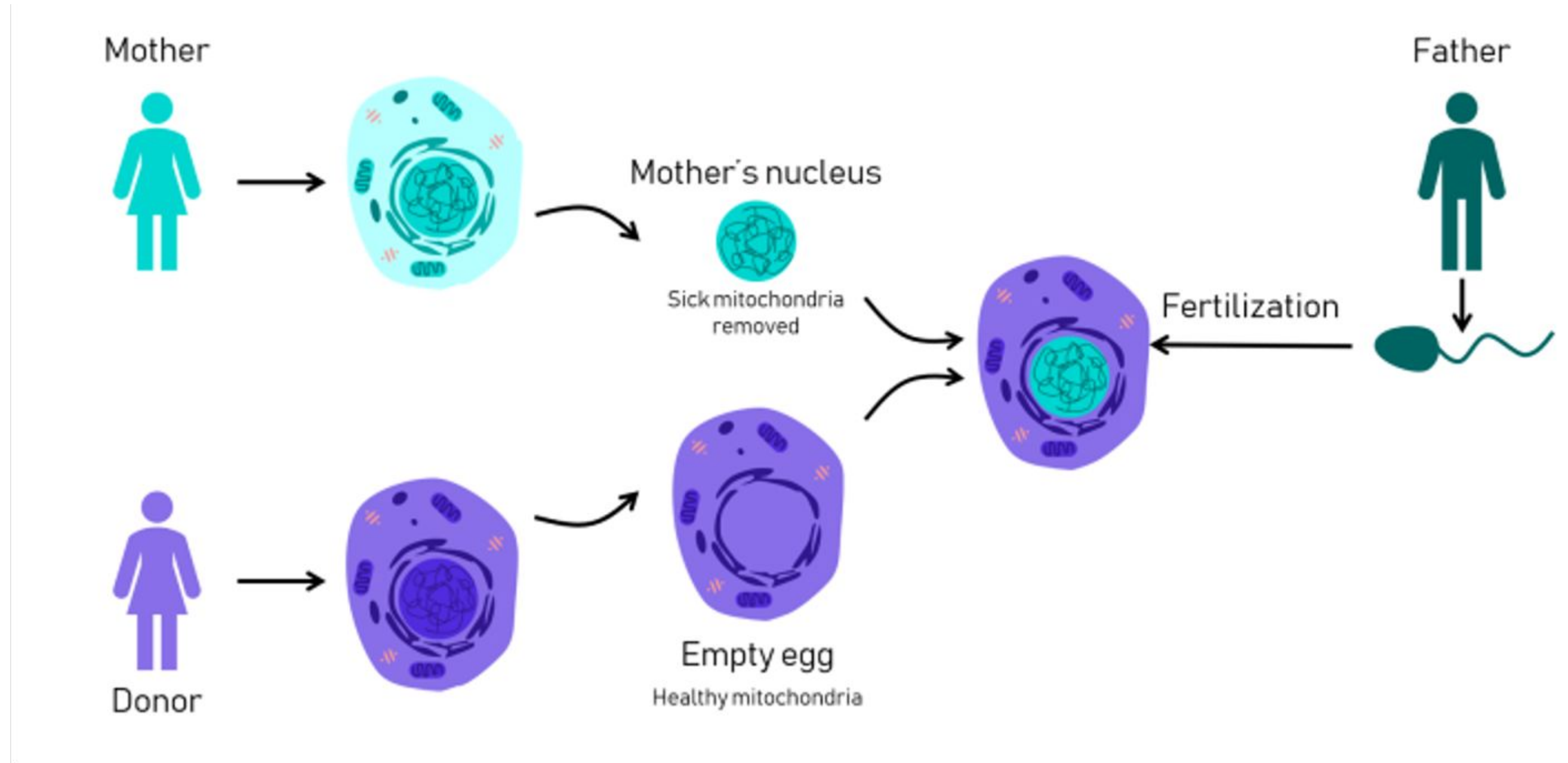


Pronuclear Transfer

After Fertilisation

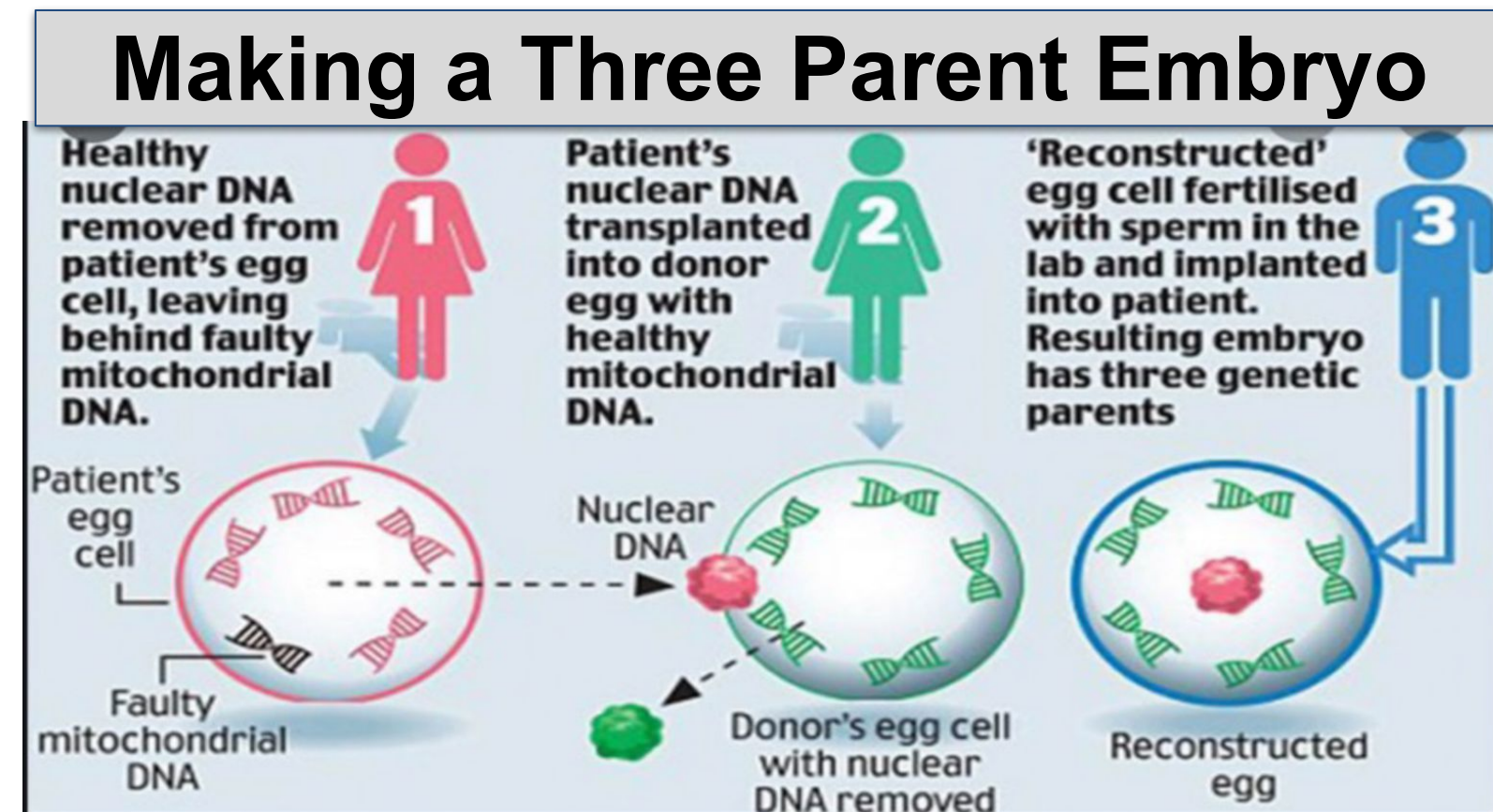
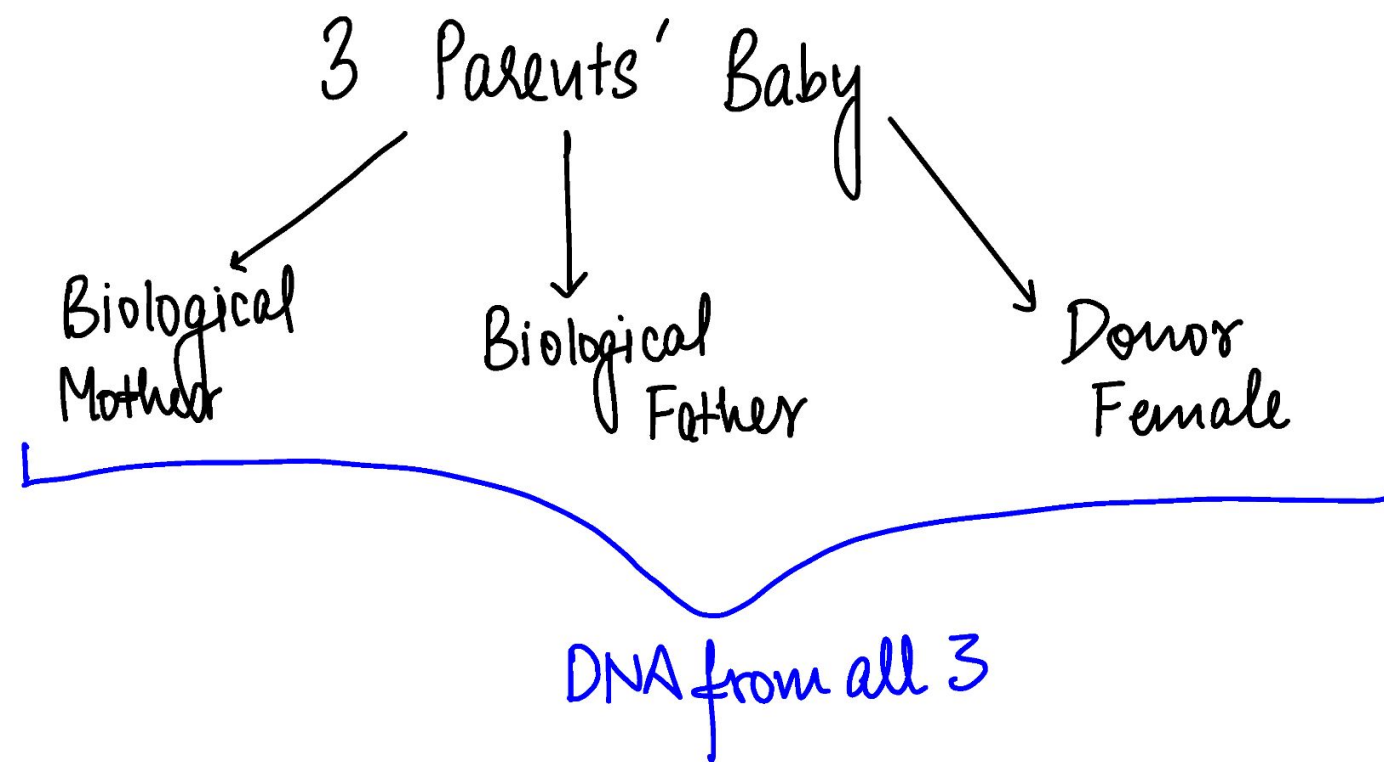
Method - 2





Reconstructed Egg Technology!

Before Fertilisation



Approved
in
UK,
Mexico.

Test of 6 Questions!

Cut Off: 6/6



Question 1

Q. In the context of hereditary diseases, consider the following statements:

1. Passing on mitochondrial diseases from parent to child can be prevented by mitochondrial replacement therapy either before or after in vitro fertilization of the egg.
2. A child inherits mitochondrial diseases entirely from mother and not from father.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Question 2

Q. In the context of recent advances in human reproductive technology, “Pronuclear Transfer” is used for:

- a) fertilization of egg in vitro by the donor sperm
- b) genetic modification of sperm producing cells
- c) development of stem cells into functional embryos
- d) prevention of mitochondrial diseases in offspring

Question 3

Q.7) Which of the following statements are correct regarding the general difference between plant and animal cells?

1. Plant cells have cellulose cell walls whilst animal cells do not.
2. Plant cells do not have plasma membrane unlike animal cells which do.
3. Mature plant cell has one large vacuole whilst animal cell has many small vacuoles.

Select the correct answer using the code given below:

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Question 4

Q.Which of the following cell organelles play the most significant role in protein synthesis?

- A.Lysosome and Centrosome
- B.Endoplasmic reticulum and Ribosome
- C.Golgi apparatus and Mitochondria
- D.Lysosome and Mitochondria

Question 5

Q. Which organelle in the cell, other than the nucleus, contains DNA?

- (a) Centriole
- (b) Golgi apparatus
- (c) Lysosome
- (d) Mitochondrion

Question 6

Q. Consider the following statements: DNA Barcoding can be a tool to:

1. assess the age of a plant or animal.
2. distinguish among species that look alike.
3. Identify undesirable animal or plant materials in processed foods.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 3 only
- (c) 1 and 2
- (d) 2 and 3

Congratulations on giving the test!

Cut-Off Figure —→ Only for Pressure

Let's Discuss The Answers.



Question 1 (Year 2021)

Q. In the context of hereditary diseases, consider the following statements:

1. Passing on mitochondrial diseases from parent to child can be prevented by mitochondrial replacement therapy either before or after in vitro fertilization of the egg.
2. A child inherits mitochondrial diseases entirely from mother and not from father.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2**
- d) Neither 1 nor 2

Answer: C

- The technological development allows nearly complete replacement of the cytoplasm of egg/embryo, eliminating the transmission of undesired defective mitochondria for patients with inherited mitochondrial diseases, which is called mitochondrial replacement therapy (MRT). These processes use assisted reproductive technology, including in vitro fertilization. The father's sperm can fertilize the patient egg before the maternal genetic material is removed or after the maternal genetic material has been transferred to the donor egg. Hence, statement 1 is correct.
- Mitochondrial diseases are chronic (long-term), genetic, often inherited disorders that occur when mitochondria fail to produce enough energy for the body to function properly. As the mitochondrial genome is inherited exclusively from the mother, the child inherits mitochondrial diseases entirely from mother and not from father. Hence, statement 2 is correct.
- Therefore, option (c) is the correct answer.

Question 2 (Year 2020)

Q. In the context of recent advances in human reproductive technology, “Pronuclear Transfer” is used for:

- a) fertilization of egg in vitro by the donor sperm
- b) genetic modification of sperm producing cells
- c) development of stem cells into functional embryos
- d) prevention of mitochondrial diseases in offspring**

Answer: D

- Pronuclear transfer involves the transfer of pronuclei from one zygote to another. This technique first requires fertilisation of healthy donated egg/s (provided by the mitochondrial donor) with the intending male parent sperm. Simultaneously, the intending mother's affected oocytes are fertilised with the intending father's sperm. Both sets of fertilised oocytes are allowed to develop to the early zygote stage where the pronuclei are visible.
- Pronuclear transfer has the potential to 'treat' human mtDNA disease at a genetic level. D is correct.
- Mutations in mtDNA are a cause of mitochondrial disease, a heterogeneous group of diseases that can lead to premature death, sometimes in infancy or childhood. Most mitochondrial diseases lack specific treatments, and women who carry the causative mutations are at high risk of transmitting the diseases to their offspring. Risk of transmission is greatest for women with high heteroplasmy—women whose total mtDNA content in affected cells or tissues is made up of between 60 and 90 percent mutated mtDNA.

Question 3 (Year 2020)

Q. Which of the following statements are correct regarding the general difference between plant and animal cells?

1. Plant cells have cellulose cell walls whilst animal cells do not.
2. Plant cells do not have plasma membrane unlike animal cells which do.
3. Mature plant cell has one large vacuole whilst animal cell has many small vacuoles.

Select the correct answer using the code given below:

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only**
- d) 1, 2 and 3

Answer: C

- Plant and animal cells are different as the former possess cell walls, chloroplasts, plastids and a large central vacuole which are absent in animal cells. Hence, statement 1 and 3 are correct.
- On the other hand, animal cells have centrioles which are absent in almost all plant cells. Both plant and animal cells have plasma membranes. Hence, statement 2 is not correct.
- Therefore, option (c) is the correct answer.

Question 4 (Year 2001)

Q.Which of the following cell organelles play the most significant role in protein synthesis?

A.Lysosome and Centrosome

B.Endoplasmic reticulum and Ribosome

C.Golgi apparatus and Mitochondria

D.Lysosome and Mitochondria

Answer: B

As explained in the class.

Question 5 (Year 2001)

Q. Which organelle in the cell, other than the nucleus, contains DNA?

- (a) Centriole
- (b) Golgi apparatus
- (c) Lysosome
- (d) Mitochondrion**

Answer: D

As explained in the class.

Question 6 (Year 2022)

Q. Consider the following statements: DNA Barcoding can be a tool to:

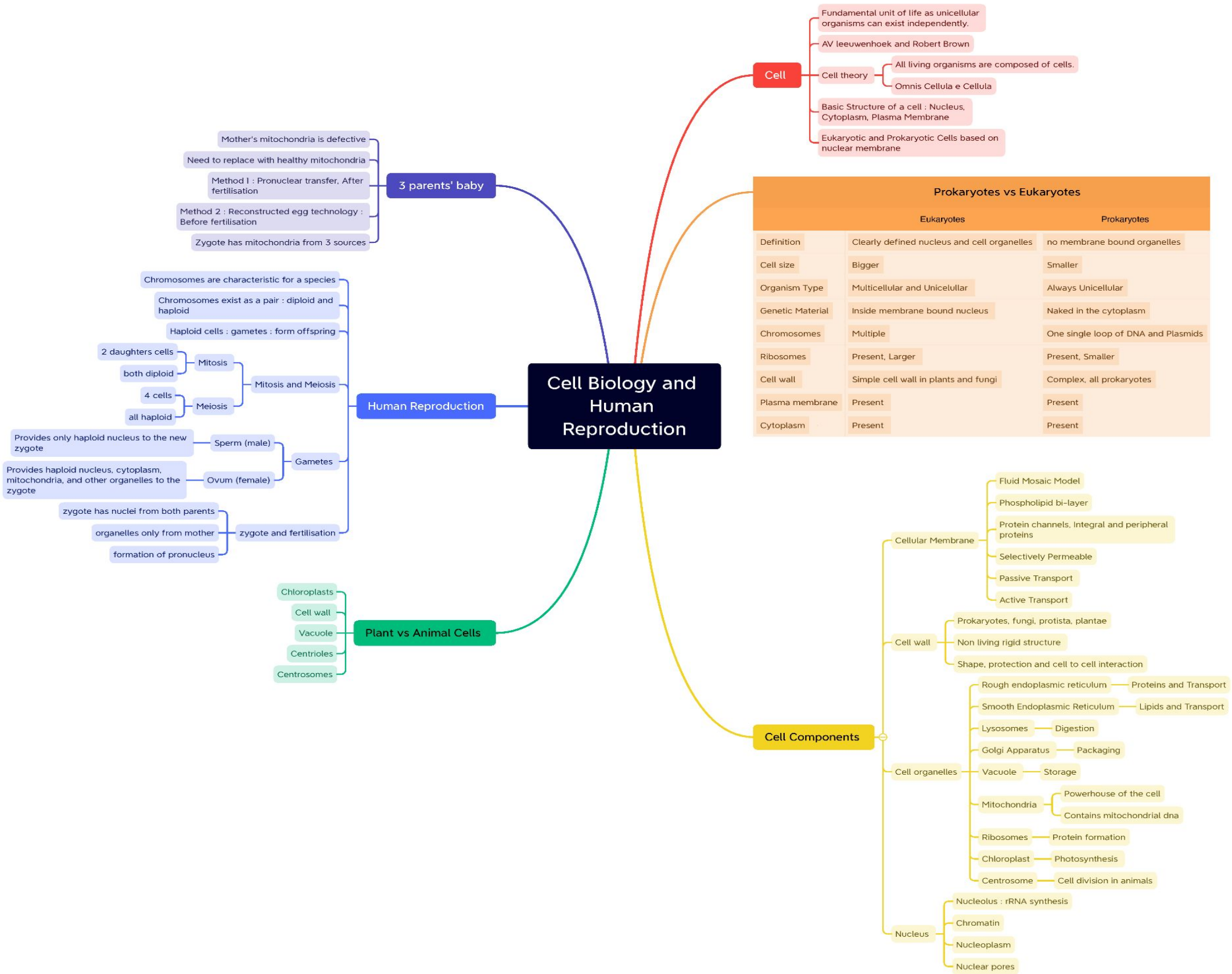
1. assess the age of a plant or animal.
2. distinguish among species that look alike.
3. Identify undesirable animal or plant materials in processed foods.

Which of the statements given above is/are correct?

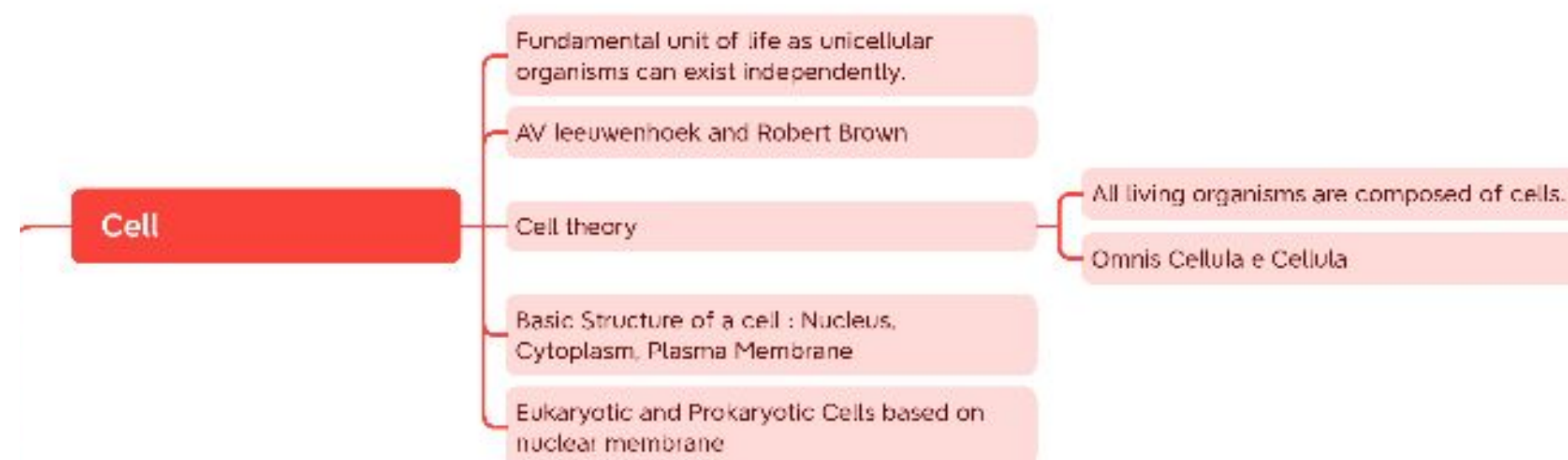
- (a) 1 only
- (b) 3 only
- (c) 1 and 2
- (d) 2 and 3**

Answer: D

- DNA barcoding is a system for species identification focused on the use of a short, standardized genetic region acting as a “barcode”
- DNA barcoding has got no association with determining the age of an organism.
- DNA barcoding is a system for fast and accurate species identification that makes ecological system more accessible by using short DNA sequences instead of whole genome and is used for eukaryotes.
- In recent times, DNA barcoding has emerged as the most potent tool for detection of adulteration of food with unwanted plant and animal material.



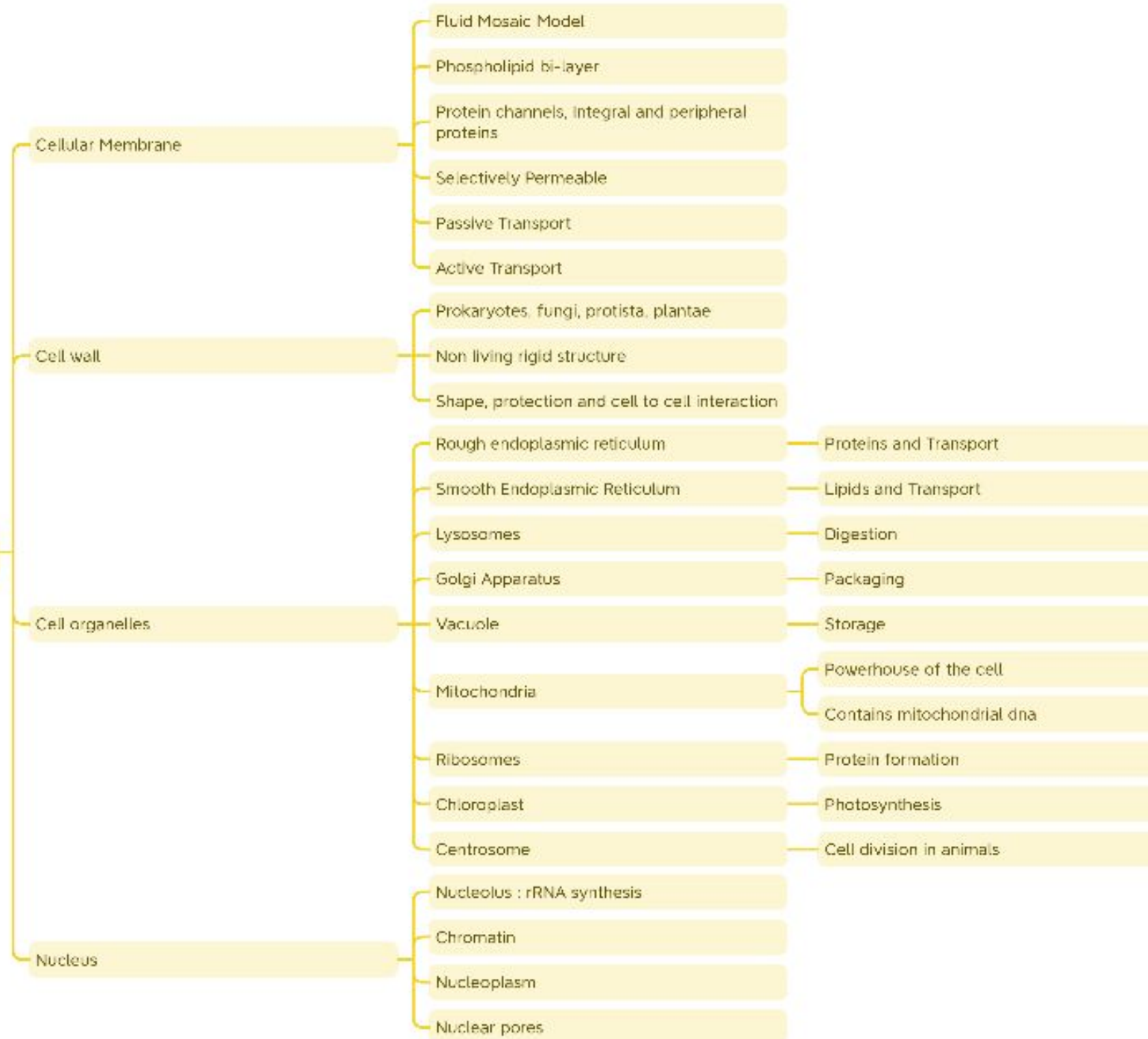
Cell Biology and Human Reproduction



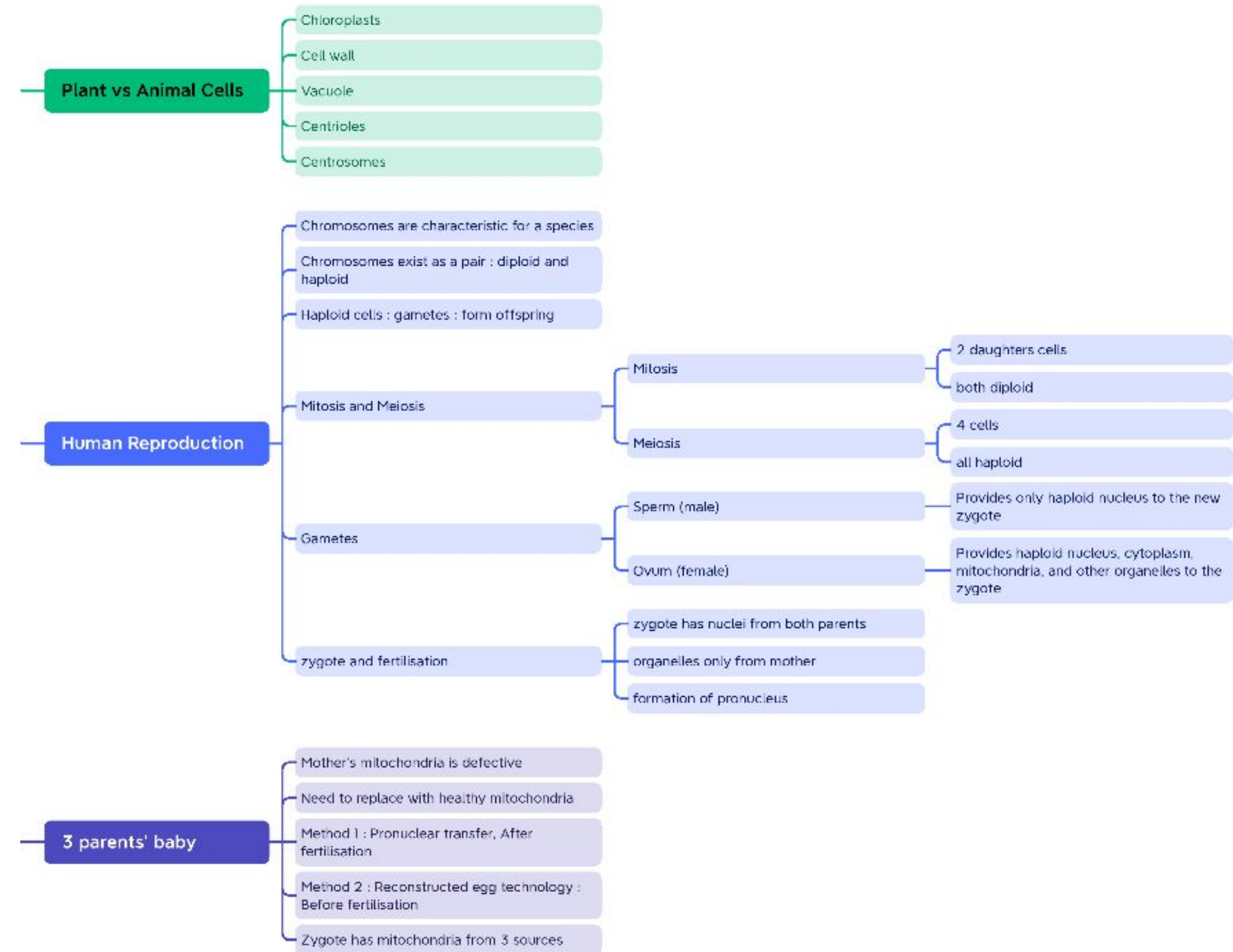
Prokaryotes vs Eukaryotes		
	Eukaryotes	Prokaryotes
Definition	Clearly defined nucleus and cell organelles	no membrane bound organelles
Cell size	Bigger	Smaller
Organism Type	Multicellular and Unicellular	Always Unicellular
Genetic Material	Inside membrane bound nucleus	Naked in the cytoplasm
Chromosomes	Multiple	One single loop of DNA and Plasmids
Ribosomes	Present, Larger	Present, Smaller
Cell wall	Simple cell wall in plants and fungi	Complex, all prokaryotes
Plasma membrane	Present	Present
Cytoplasm	Present	Present

Cell Biology and Human Reproduction

Cell Components



Cell Biology and Human Reproduction



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